Cat® Energy Control System (ECS)

Cat ECS SCADA Modbus Software Manual

Cat® ECS 100 / Cat ECS 200

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Table of Contents

TABLE OF CONTENTS *3*

FOREWORD *6*

1.0 GENERAL INFORMATION *7*

1.1 INTRODUCTION 7

1.2 APPLICATIONS 8

1.3 REFERENCES 8

2.0 SAFETY *9*

2.1 ELECTRICAL SAFETY 9

3.0 GLOSSARY OF TERMS *10*

4.0 CAT ECS SCADA MODBUS OVERVIEW *13*

5.0 SCADA DATA LINK SETUP AND CONFIGURATION *14*

5.1 SCADA DATA LINK CONFIGURATION 14

5.1.1 SCADA IP Configuration Setup via Cat ET 14

5.1.2 SCADA Data Link Enabling and Password Setup via Cat ET 15

5.1.3 SCADA Configuration Setup via HMI Display 15

5.1.4 SCADA Data Link Access Password Specifics 17

6.0 CAT ECS SCADA MODBUS DATA INTERPRETATION AND SCALING *18*

6.1 NUMERICAL DATA 18

6.1.1 Numerical Data Read – One (1) Word Unsigned Integer 18

6.1.2 Numerical Data Write – One (1) Word Unsigned Integer 18

6.1.3 Numerical Data Read – Two (2) Word Unsigned Integer 19

6.1.4 Numerical Data Write – Two (2) Word Unsigned Integer 20

6.1.5 Numerical Data - Byte Extracting 21

6.1.6 Numerical Data - Byte Packing 22

6.2 STATE / STATUS / COMMAND / INFORMATION DATA 23

6.2.1 State / Status / Info Data 23

6.2.2 Command Data 23

6.3 COMPLEX DATA 23

7.0 CAT ECS MODBUS DATA *24*

7.1 GENSET 25

7.1.1 Genset Rating (Active Group) 28

7.1.2 Genset AC 28

7.1.3 Genset Power 29

7.1.4 Genset Energy 29

7.1.5 State, Mode and Status 30

7.1.6 Genset Start / Stop 31

7.1.7 Battery Voltage 31

7.1.8 Base Load, Power Factor and Reactive Load (Write Commands) 32

7.1.9 Load Sense Load Demand 32

7.1.10 Load Share and Load Control 34

7.1.11 Utility Transfer (Write Command) 34

7.1.12 Generator Winding and Bearing Temperature 35

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7.1.13 Gas Substitution (Write Command) ... 35
7.1.14 Generator Paralleling Data (Gen #1 to Gen #32) ... 36
7.2 ENGINE ... 38
7.2.1 Service Meter ... 38
7.2.2 Engine Speed ... 38
7.2.3 Engine Fuel ... 38
7.2.4 Engine Oil ... 39
7.2.5 Engine Coolant ... 39
7.2.6 Engine Exhaust ... 40
7.2.7 Engine Intake ... 40
7.2.8 Engine Turbocharger ... 40
7.2.9 Engine Filter ... 41
7.2.10 Engine Aftertreatment ... 41
7.2.11 Engine Gas ... 43
7.2.12 Engine Dynamic Gas Blending ... 45
7.2.13 Engine – Other Data ... 45
7.3 PROGRAMMABLE CYCLE TIMER ... 46
7.4 REALTIME CLOCK ... 47
7.4.1 Realtime Clock Data Consist ... 47
7.5 CAT ECS INPUTS AND OUTPUTS ... 48
7.5.1 Read - Onboard Inputs and Outputs ... 48
7.5.2 Write - Onboard Outputs ... 49
7.5.3 Read - Remote Inputs and Outputs ... 49
7.5.4 Write - Remote Outputs ... 51
8.0 CAT ECS EVENTS ... 52
8.1 ACKNOWLEDGE AND RESET ... 52
8.2 ACKNOWLEDGE STATUS ... 52
8.3 HARD SHUTDOWN, SOFT SHUTDOWN, LAMP, HORN AND DISPLAY EVENTS ... 52
8.4 USER ALARM GROUPS (SCADA ALARM BIT PACKING) ... 53
8.5 ANNUNCIATOR ... 54
8.6 EVENT LOGS ... 55
8.6.1 Event Logs – Number of Active Events ... 55
8.6.2 Event Logs – System Events (Genset and Engine) ... 56
8.6.3 Event Logs – Status Events ... 56
9.0 NETWORK AND MODULE ... 57
9.1 NETWORK STATUS ... 57
9.2 MODULE ONLINE STATUS ... 57
9.3 MODULE INFORMATION ... 58
10.0 GRID CODE ... 60
11.0 BALANCE OF PLANT ... 66
12.0 EXAMPLES ... 68
12.1 SCADA PASSWORD WRITE ... 68
12.2 ENGINE SPEED MONITOR AND CONTROL ... 69
12.3 GENSET VOLTAGE MONITOR AND CONTROL ... 71
12.4 BASE LOAD (kW) MONITOR AND CONTROL ... 74

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12.5 Analog Inputs and Outputs **76**
12.5.1 Read - Analog Input and Output 76
12.5.2 Write - Analog Output 76
12.5.3 Read - PWM Output 78
12.5.4 Write - PWM Output 78
12.6 Remote Analog Inputs and Outputs **80**
12.6.1 Read - Remote Analog Input and Output 80
12.6.2 Write - Remote Analog Output 80
12.6.3 Read - Remote RTD Input 81
12.7 Event Logs **82**
13.0 BEST PRACTICES ***86***
13.1 Data and Network Optimization **86**
13.2 Retry Mechanisms and Error Handling **86**
13.3 Security Considerations **86**
14.0 TROUBLESHOOTING ***87***
14.1 Intermittent or Failed Communication **87**
14.2 Modbus Errors - Exception Responses **88**
14.3 Invalid or Out of Range Data **89**

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Foreword

This manual facilitates user Modbus client software integration with Cat® ECS family of genset controllers Cat ECS 100 and Cat ECS 200 over the Cat ECS SCADA Modbus interface.

The Modbus data in this document is grouped into sections and subsections according to its context such as Genset, Engine, Inputs, Outputs and Events. This grouping and data contextualization helps user in faster software integration and deployment time.

To ease software implementation, illustrations and examples are presented using demonstrations which eventually lead to predictable software integration timeline. These include conversion of single and two-word numeric data, byte-data extraction and packing, engine speed monitoring and control, voltage monitoring and control, Base Load Monitoring and Control, Onboard and Remote Analog Inputs and Outputs read and write.

Best practices are provided in this document to help user to achieve reliable and efficient data exchange. These best practices not only minimize communication intermittences and failures but help user achieving predictable performance and yield enhanced performance in large control networks.

In various places throughout this document, references are made to the other manuals and documentation. Other product specific information and data is available from a variety of sources. For more information, see the reference section or contact the Cat dealer or dealer technical communicator (TC) nearest you.

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1.0 General Information

1.1 Introduction

Modbus is a widely used industrial communication protocol due to its availability in majority of the industrial controllers (PLCs, HMIs and SCADA systems). Modbus TCP network consists of Modbus servers and Modbus clients. Modbus clients initiate Read and Write requests to Modbus servers, and Modbus servers respond in a well-defined manner.

Each Cat ECS controller (Cat ECS 100 Genset, Cat ECS 200 Genset and Cat ECS 200 Paralleling) has three (3) sets of Modbus servers; one of READ (TCP Port 502), one WRTIE (TCP Port 50200) and one for EVENTS (TCP Port 50201). Modbus TCP client PLC or SCADA system sends READ and WRITE commands to the Cat ECS controller using the controller SCADA IP address and TCP Ports. Each Cat ECS Modbus server supports up to a maximum of 5 clients. Each Modbus client can make a maximum of one request every 100ms. A Modbus exception code 6 (Slave/Server Device Busy) will be generated if the maximum number of requests or polling rates are exceeded or violated.

Modbus client PLC or SCADA is required to connect to the Cat ECS Genset Controller RJ-45 Ethernet Port #2 in Cat ECS 100 and Cat ECS 200 controls as shown in Figure 1.1. Cat ECS 200 Paralleling Controller transmits and receives Modbus data via the Genset Controller over the packaged installed 2-wire ethernet communication bus.

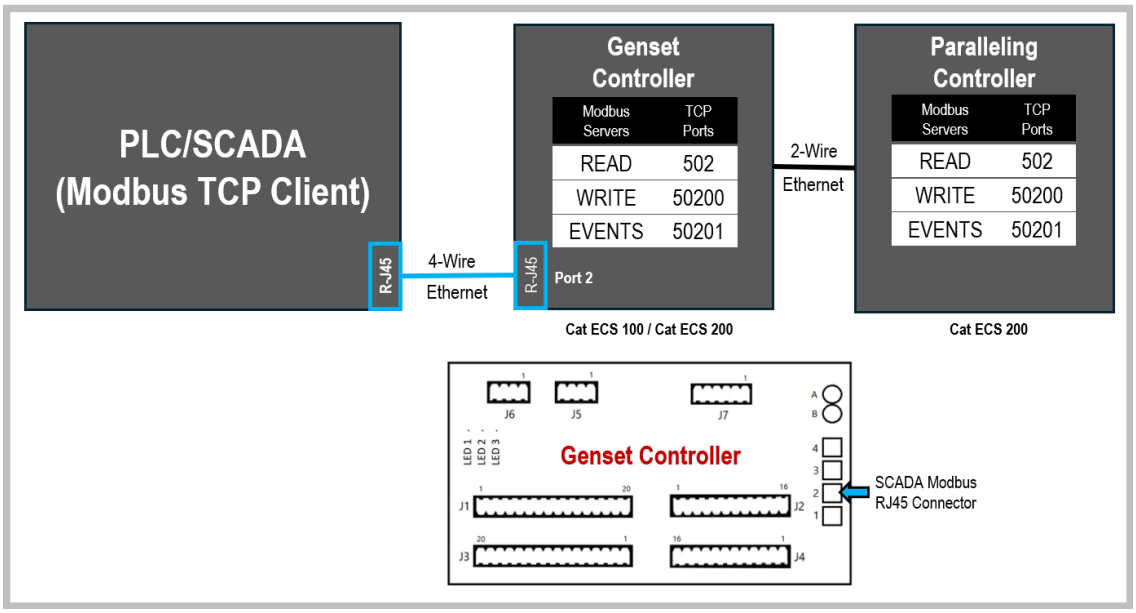

Figure 1.1: Cat ECS SCADA Modbus Interface and Customer Connection

Note: Refer "Cat ECS Communications Application & Installation Guide" for communication interfaces details.

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1.2 Applications

This guide is intended to assist users in developing Modbus client software for integration with Cat ECS 100 and Cat ECS 200 controllers. It will be updated periodically to reflect any necessary revisions.

1.3 References

The following documents and references contain important information to be consulted:

- Cat Energy Control System (ECS) Communications Application & Installation Guide
- Cat Energy Control System (Cat ECS 100 / Cat ECS 200) Modbus Registers Spreadsheets
- Cat Energy Control System (Cat ECS 100 / Cat ECS 200) Software Operations Manual
- Cat Energy Control System (Cat ECS 100 / Cat ECS 200) HMI Manual
- Cat Energy Control System (ECS) Hardware Manual
- Cat Energy Control System (Cat ECS 100 / Cat ECS 200) Service and Troubleshooting Manual
- Expansion Module Manual (DIO, RTD, TC, Annunciators, Remote Ethernet modules)
- CDVR Manual
- Detailed Wiring Diagrams
- Modbus Application Protocol Specifications at www.modbus.org

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2.0 Safety

2.1 Electrical Safety

DO NOT operate or work on a generator set unless you have read and understand the instructions and warnings in the “Operation and Maintenance Manual”. Failure to follow the instructions or heed the warnings could result in injury or death. Contact any Cat® dealer for replacement manuals. Proper care is your responsibility.

Figure: 2.1 Safety Logo

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3.0 Glossary of Terms

100G - Not a Caterpillar branded name. Used in this document only to refer Cat ECS 100 Genset controller.

200G - Not a Caterpillar branded name. Used in this document only to refer Cat ECS 200 Genset controller.

200P - Not a Caterpillar branded name. Used in this document only to refer Cat ECS 200 Paralleling controller.

A - Ampere (Current Unit)

APCDL - Advanced Parallel Control Data Link

ASCII - American Standards Code for Information Interchange

CAN - Controller Area Network.

Cat® ECS - Cat® Energy Control System (Marketing name for the control system).

Cat ET - Cat Electronic Technician (Service Tool)

CID - Component Identifier

Data Link – Refers to Ethernet or CAN bus.

DBA - Dead Bus Arbitration

DUINT - Double Unsigned Integer

ECM - Engine Control Module.

EID - Event Identifier

FMI - Failure Mode Identifier.

GSC - Generator Set Controller.

HMI - Human Machine Interface.

Hz - Hertz (Frequency Unit)

I/O - Input/Output.

INT - Integer

IP - Internet Protocol.

IVR - Integrated Voltage Regulator.

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J1939 - An application layer messaging protocol that provides client/server communications between devices connected to different types of buses or networks.

kVA - Kilo-Volt-Ampere

kVAr - Kilo Volt Ampere Reactive

kVArh - Kilo Volt Ampere Reactive Hour

kW - Kilo-Watt

kWh - Kilo Watt Hour

LSB - Least Significant Byte

LSLD - Load Sense Load Demand

LSW - Least Significant Word

MGDL - multi Genset Data Link

Modbus TCP - An application layer messaging protocol that provides client/server communications between devices connected to different types of buses or networks.

MSB - Most Significant Byte

MSW - Most Significant Word

NFPA - National Fire Protection Association. An American standards body responsible for creation of various codes.

Offset - After scaling, an offset value is added to the result. This shifts the scaled data up or down, positioning it within the desired output range.

PDU - Protocol Data Unit

PF - Power Factor

PLC - Programmable Logic Controller.

PWM - Pulse Width Modulation

Range - This is the initial and final interval of the data. Scaling adjusts the size of the range, while offsetting positions the range.

RTD - Resistance Temperature Detector.

SCADA - Supervisory Control and Data Acquisition.

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Scaling - This involves multiplying the raw input data by a scale factor. The scale factor determines how much the data is stretched or compressed.

SPN - Suspect Parameter Number. The CAN term for any parameter whose data is transmitted over a CAN network, such as the controller primary or accessory data link. For a complete list of SPNs supported by the HMI, refer to the “Cat ECS 200 Service and Troubleshooting Manual”.

UINT - Unsigned Integer

ULONG - Unsigned Long

V - Volt (Voltage Unit)

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4.0 Cat ECS SCADA Modbus Overview

The Cat ECS SCADA Modbus uses holding registers. Each register consists of 16-bit long one (1) word register.

The Cat ECS SCADA Modbus register addresses in this document are normally referenced by the name and the decimal address. However, there is a different addressing model called "PDU Addressing" in which Modbus addresses for holding registers are in the range 40,000-49,999. For applications using PDU Addressing, simply add 40,000 to the addresses given in this document. Modbus addresses given in this document and the Modbus register spreadsheets are different from the address numbers transmitted over the data link. The address of 1 is the lowest address but is transmitted as 0x0000. Therefore, when register addresses are given in hexadecimal, note that the decimal value must be decreased by 1 and then converted to hex or decimal.

SUPPORTED FUNCTION CODES

The Cat ECS SCADA Modbus supports the following Modbus Function Codes:

- **Function code 3 (0x03)** – Read Registers
- **Function code 6 (0x06)** – Write Single Register
- **Function code 16 (0x10)** – Write Multiple Registers

Note: The usage of Cat ECS SCADA Modbus registers is also supported in the Cat ECS Integrated PLC Feature Set. Refer to Cat ECS 100 and Cat ECS 200 Software Operating Manuals Section "**Integrated PLC Feature Set"** for further information.

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5.0 SCADA Data Link Setup and Configuration

5.1 SCADA Data Link Configuration

Cat ECS SCADA Modbus data link requires initial configuration setup. This configuration setup can be done via Cat ET or HMI.

5.1.1 SCADA IP Configuration Setup via Cat ET

- Connect to the Cat ECS controller with communication adapter via Cat ET. Open the Service Tab at the top and select "IP Address Configuration".
    - **Cat ECS 100 -** Select "SCADA IP" from the list and then configure the Genset Control SCADA IP address with other parameters according to the site-specific network IP configuration.
    - **Cat ECS 200 -** Connect the Genset and Paralleling controllers at a time using communication adapter via Cat® ET. Select "SCADA IP" from the list and then configure the SCADA IP addresses of the two controllers one at a time with other parameters according to the site-specific network IP configuration.

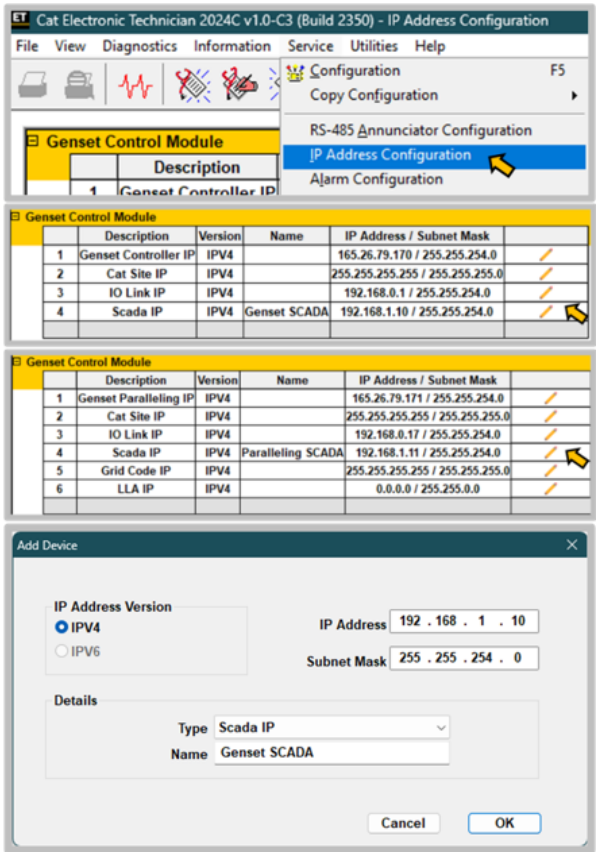

Figure 5.1.1: Cat ET IP Configuration

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5.1.2 SCADA Data Link Enabling and Password Setup via Cat ET

Cat ECS SCADA Data Link requires three (3) main setpoints to configure. The Diagnostic setpoints are optional and can be configured accordingly.

Ethernet	
SCADA Port Enable Status	Enabled
SCADA Data Link Access Password	00000000
SCADA Data Link Remote Control Enable Status	Enabled
TCP/IP SCADA Data Link Diagnostic Response Configuration	Disable Condition
TCP/IP SCADA Data Link Diagnostic Audible Alert	No
TCP/IP SCADA Data Link Diagnostic Breaker #1 Trip	No
TCP/IP SCADA Data Link Diagnostic Breaker #2 Trip	No
TCP/IP SCADA Data Link Diagnostic Fault Protection Timer	No

Figure 5.1.2: Cat ET Cat ECS SCADA Enable and Password Setup

1. **SCADA Port Enable Status**
   Used to read and program the enable status for supervisory control and data acquisition (SCADA) data link. If set to "Disabled", no read or write of the SCADA MODBUS registers is available.
2. **SCADA Data Link Remote Control Enable Status**
   This setpoint determines if SCADA MODBUS writes will be allowed on the controller. If this is "Disabled", Write commands from the Modbus client will not be allowed. This setpoint only effects the SCADA Modbus writes and does not affect Modbus Read and external displays on the Cat network.
3. **SCADA Data Link Access Password**
   SCADA Data Link Access Password is an eight (8) word, or sixteen (16) bytes long ASCII password entered to the Cat ECS 100 / Cat ECS 200 genset controller via Cat ET or HMI. Password is empty by default and must be set to a value to write values or commands to Cat ECS controller over the SCADA Modbus.

5.1.3 SCADA Configuration Setup via HMI Display

Navigate to Menu à Configure à SCADA settings and then configure the required setpoints explained in above section.

HMI Display – Access Level Specifics:

- The HMI display requires **Access Level 2 (Technician)** to configure the following:
  - SCADA password
  - SCADA IP address
  - IP Network Mask
- The HMI display requires **Access Level 1 (Operator)** to configure the following:
  - SCADA data link Enable Status
  - SCADA data link Remote Control Status

Note: Refer "Cat Energy Control System (Cat ECS 100 / Cat ECS 200) HMI Manuals" for further information.

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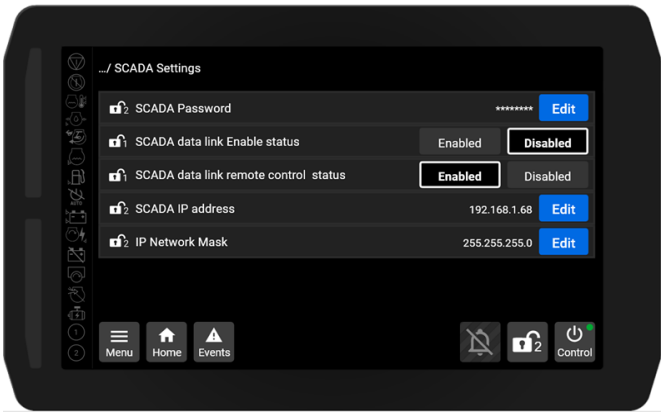

Figure 5.1.3: Cat ECS 100 HMI Display – SCADA Setup Configuration

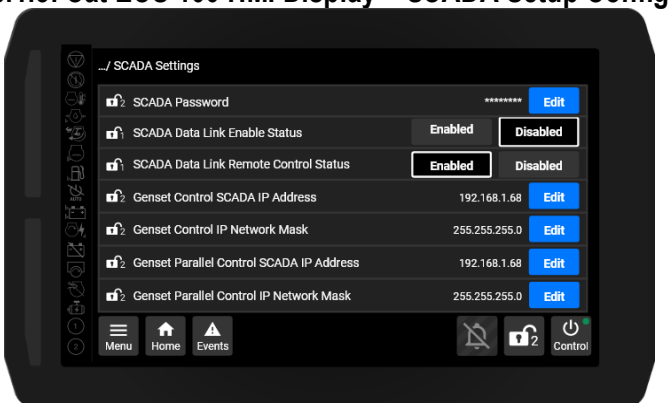

Figure 5.1.4: Cat ECS 200 HMI Display - SCADA Setup Configuration

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5.1.4 SCADA Data Link Access Password Specifics

- **SCADA Password Data Type and Format**:
  - SCADA password is stored as ASCII in the Cat ECS genset controller, with each ASCII character taking one (1) byte. Cat ECS SCADA Modbus supports alpha numeric values for the password.
  - SCADA Data Link Password is Four (4) words, or sixteen (8) bytes long.
  - Remaining unused words or bytes of the password value must be set to ASCII space (SP). The first space character (SP) terminates the string, so an entry of 12_3 (space between the 2 and the 3) will return to an exception error since only spaces (or nulls, after the first 8 bytes) are accepted after the first space.
  - Leading zeros get dropped off, and trailing zeros are kept. For example, 01 is stored as 1, while 10 is stored as 10.
- **SCADA Password Values and Write Access:**
  - For all eight (8) zero values password (**00000000**), Modbus client does not require a password to write values or commands to Cat ECS controller.
  - For non-zero password values, Modbus client must require password to write to Cat ECS controller.
- **SCADA Data Link Inactivity Timer (30 seconds) and Write Access:**
  - SCADA data link has a fixed internal Inactivity timer which activates and times out after 30 seconds of no data link activity. This inactivity timer applies only when the password is set to non-zero values. Modbus server responds an exception error of 0x**FF FF** xx xx (or decimal **255 255** xx xx and the Modbus client will require to send the SCADA Data Link Access Password to the Cat ECS controller to re-establish the write access.
  - The causes of non-activity can be related to any of the following:
    - Cat ECS controller powered off
    - Ethernet cable disconnected
    - Modbus TCP client goes offline
    - Modbus TCP client Stops reading or writing
    - Any Other Cause.

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6.0 Cat ECS SCADA Modbus Data Interpretation and Scaling

Cat ECS uses unsigned integers for both read and write Modbus data. Depending on the PLC or SCADA system data types; for example, UINT (Unsigned Integer) data type is used for 16-bit unsigned integer word and DUINT (Double Unsigned Integer) data type is used for 32-bit unsigned two-word integer where the data spans to two (2) 16-bit register words. The negative numerical data values are calculated by doing conversion and scaling on the unsigned integer value. The Cat ECS read and write data consists of:

- Numerical Data
- Status Data
- State Data
- Information Data
- Complex Data

This data needs conversion for interpretation according to the scaling, offset and status / state/ info explained in the following sections.

6.1 Numerical Data

Cat ECS SCADA Modbus numerical data consists of either Single word (16-bits / 2-bytes) or Double word (32-bits / 4-bytes) unsigned integers.

Cat ECS SCADA numerical Modbus data uses Big-Endian format for the data that spans over two (2) words. In Big-Endian format, MSW (Most Significant Word) value of the two-word data is placed at the lower Modbus register address and LSW (Least Significant Word) value of the two-word data is placed at the higher Modbus registers for Modbus reads and writes.

Numeric data conversion of two-word and byte in the following sections is demonstrated by using Microsoft Windows Calculator to facilitate PLC/SCADA software implementation and troubleshooting. User can use the standard functions and operators according to their PLC or SCADA system for Bit-Shift Left (**<<**), Bit-Shift Right (**>>**), AND, OR and ADD (**+**).

6.1.1 Numerical Data Read – One (1) Word Unsigned Integer

Multiply the register raw value by scaling and then add the offset. The range provides the engineering value range of the data point. “Read Scaled Value” will be considered invalid if out of the specified range.

One (1) Word Scaled Value (READ) = Register Raw Value x Scaling + Offset

6.1.2 Numerical Data Write – One (1) Word Unsigned Integer

Subtract offset from the required engineering value and then divide by the scaling. The range provides the engineering value range of the data point.

One (1) Word Raw Value (WRITE) = (Engineering Value - Offset) / Scaling

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6.1.3 Numerical Data Read – Two (2) Word Unsigned Integer

Steps:

1) Compute DUINT (Double Unsigned Integer) Value - Left shift MSW Word Register Value (**<<**) 16 bits then Add (**+**) LSW Register Value to it.
2) Calculate engineering value using DUINT value, Scaling Factor and Offset

Table 6.1: Two (2) Word Data Read

Description TCP Port: 502	Word Count	Register 200P	Raw Value	Calculation
Analog Input #1 Data Value	2	705 706	32128 10240	**DUINT Value: MSW << 16 + LSW** (32128) **<<** 16 = 2105540608 2105540608 **+** 10240 = 2105550848 **Scaled Value: DUINT Value x Scaling + Offset** 2105550848 x (1/256) + (-8224768) = 40

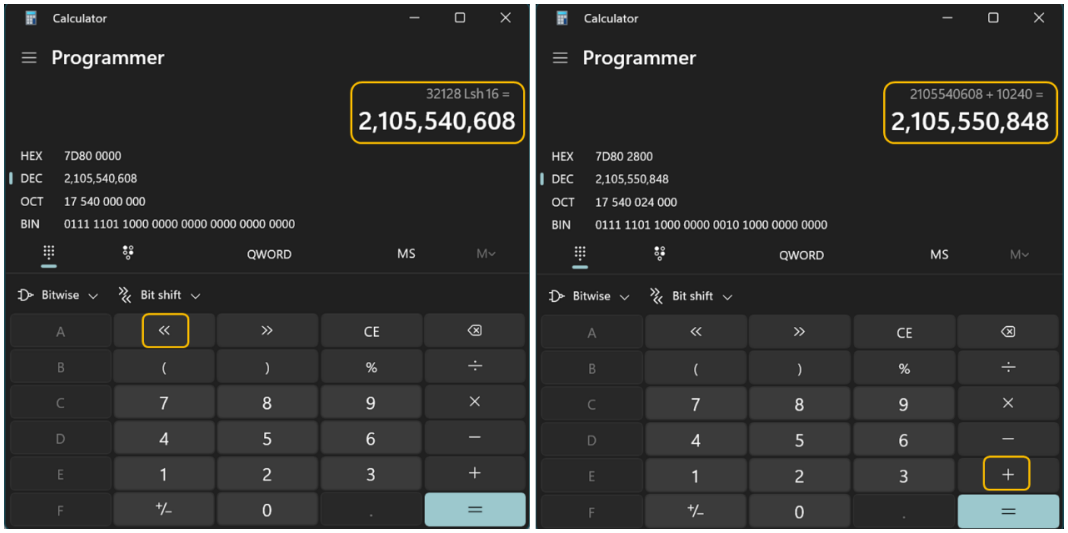

Figure 6.1: MS Calculator Demonstration - Two (2) Word Data Read

Demonstration – Microsoft Windows Calculator:

Open MS Windows calculator and select Programmer mode:

- Enter the MSW register value "32128" placed in the lower register address 705.
- Click Left Shift button (**<<**), click "**16**" to shift 16 bits to the left and then click equals (**=**)"
- Add (**+**) LSW register value "10240" placed in the higher register address 706 to the result, then click equals (**=**)
- The resultant will be the DUINT (Double Unsigned Integer)
- Now calculate engineering value using DUINT value, Scaling Factor and Offset

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6.1.4 Numerical Data Write – Two (2) Word Unsigned Integer

Steps:

1) Compute DUINT raw value by using engineering value, scaling factor and offset
2) Compute MSW 16-word write value - Right Shift (**>>**) 16 bits the computed DUINT value
3) Compute LSW 16-word write value. Perform an **AND** operation to DUINT value and 65535 (0xFFFF)

Table 6.2: Two (2) Word Data Write

Description TCP Port: 50200	Word Count	Register 200P	Raw Value	Calculation
Analog Output #1 Data Value	2	711 712	32128 768	**Analog Output (Engineering) Value:** 3 Vdc **DUINT Write Value: (Engineering Value – Offset) / Scaling =** ((3 – (-8224768) / (1/256) = 2105541376 **MSW Write Value: DUINT Value >> 16** (2105541376) **>>** 16 = 32128 **LSW Write Value: DUINT Value AND 65535** (2105541376) **AND** 65535 = 768

4) Place MSW value (32128) at the lower address 711, and LSW value (768) at the higher address (712)
5) Send write command to the Cat ECS paralleling controller TCP Port 50200.

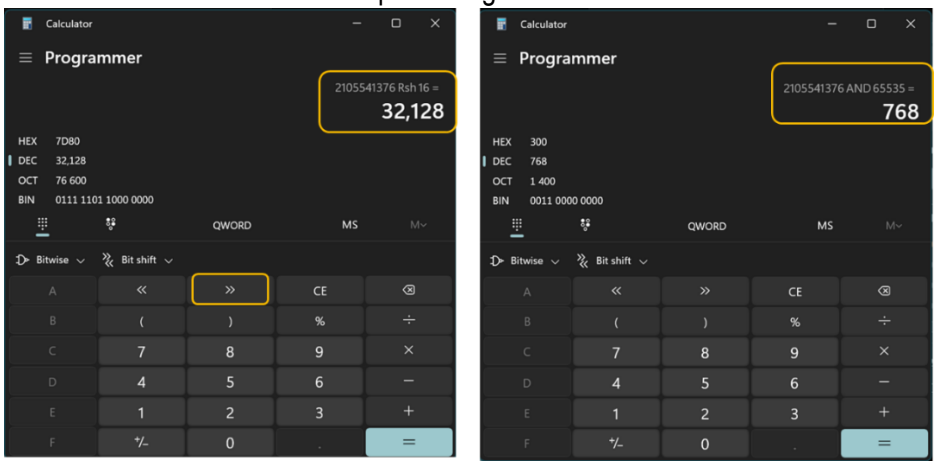

Figure 6.2: MS Calculator Demonstration - Two (2) Word Data Write

Demonstration – Microsoft Windows Calculator:

Open MS Windows calculator and select Programmer mode:

- Compute MSW write value:
  - Enter the DUINT value 2105541376.
  - Click Right Shift button "**>>**" enter "**16**" then click "**=**" to get MSW write value 32128
- Compute LSW Write value:
  - Enter the DUINT value 2105541376
  - Click **AND,** enter 65535 and then click "**=**" to get LSW write value

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6.1.5 Numerical Data - Byte Extracting

Steps:

1) Extract MSB (Most Significant Byte) – Right shift (**>>**) 8 bits the 16-bit word
2) Extract LSB (Least Significant Byte) – Perform (**AND)** operation to the 16-bit word with 255 (0xFF)

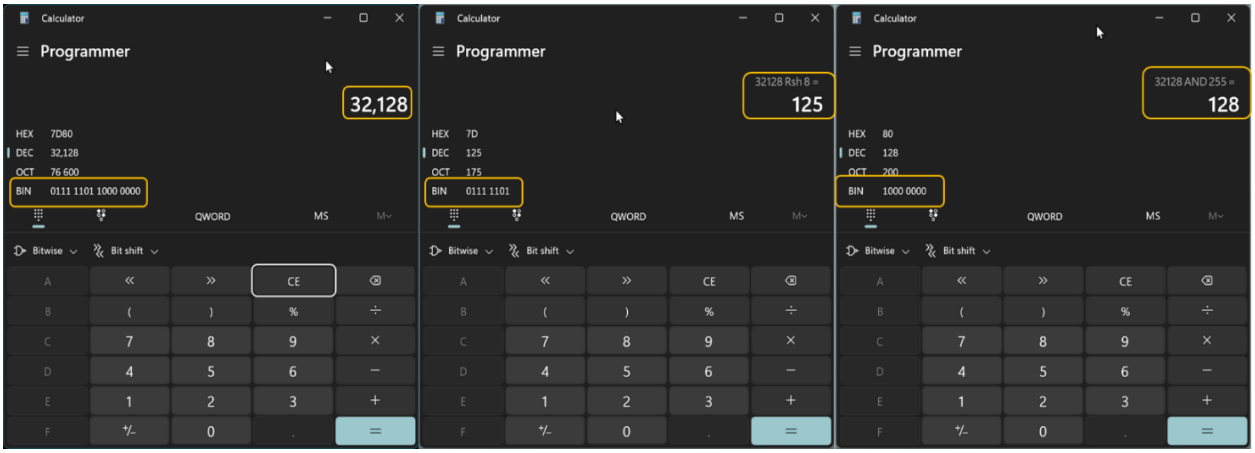

Figure 6.2: Example – Byte Extracting

Demonstration – Microsoft Windows Calculator:

Open MS Windows calculator and select Programmer mode. In this example, the 16-bit word is **32128**

- Extract MSB:
  - Enter the UINT value 32128
  - Click Right Shift button "**>>**" enter "**8**" then click "**=**".
  - MSB (Most Significant Byte) Value = **125** (0111 1101b)
- Extract LSB:
  - Enter the UINT value "32128".
  - Click (**AND),** enter 255 and then click "**=**".
  - LSB (Least Significant Byte) Value = **128 decimal** (1000 0000b).

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6.1.6 Numerical Data - Byte Packing

Steps:

- Left shift (**<<**) 8 bits the MSB (Most Significant Byte)
- Perform an (**OR)** operation with above shifted value and LSB (Least Significant Byte)

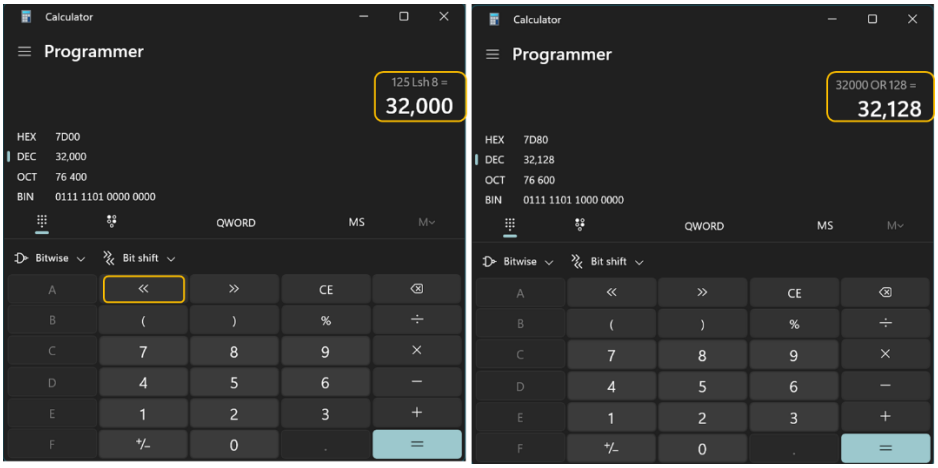

Figure: 6.3 Example – Byte Packing

Demonstration – Microsoft Windows Calculator:

Open MS Windows calculator and select Programmer mode. In this example, the MSB (Most Significant Byte) is **125** and LSB (Least Significant Byte) is **128**.

- Enter MSB value **125**
- Click Left Shift button "**<<**" enter "**8**" then click "**=**".
- Click (**OR**), enter LSB value **128**
- 16-bit Word Value = **32128**

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6.2 State / Status / Command / Information Data

6.2.1 State / Status / Info Data

There are READ data points in Cat ECS controller that are state, status or information based. These READ values are interpreted at the raw word, byte or bit level as indicated in the following sections of this document. Some of the examples are:

- Digital input and output status
- Engine Operating Mode
- Automatic Start/Stop State
- Sync Mode Switch State
- Generator Circuit Breaker Status

6.2.2 Command Data

There are WRITE data points in Cat ECS controller that are On or Off command type and require the commands number value. These values are written at the raw word level as indicated in the following sections of this document. Some of the examples are:

- Bypass Cooldown Command
- Remote Initiate Command
- Group Start Command
- Engine Control Switch Command
- Transfer to Generator Power Command

6.3 Complex Data

Complex data is an aggregate data that is made up of multiple pieces of data (Bytes and Words). Those pieces, in turn, can be numerical data, state-based data, or even complex data themselves. Some of the examples are:

- Realtime Clock
- Timestamp

The consist of these complex data made up of years, months, days, hours, minutes, and seconds.

Note: User can use Byte extraction and packing techniques explained in above section to read and write the complex data accordingly.

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7.0 Cat ECS Modbus Data

The SCADA Modbus Registers and associated details in this document are grouped into sections and subsections according to the data context and scaling conversion with respect to Cat ECS 100 and Cat ECS 200 controllers. This grouping and contextualization of data facilitates faster user software integration and reduces software deployment time.

The Modbus data tables in this manual are based on Cat ECS 100 v1.4 and Cat ECS 200 v1.1 registers list spreadsheets. These versions of the Cat ECS 100 and Cat ECS 200 Modbus register lists are current as of the time of the publication of this manual. Though a comprehensive list of SCADA Modbus registers is provided in this document to assist the user, not all the Modbus Registers are included from the Modbus Register Lists. Therefore, user is advised to refer to Cat ECS 100 v1.4 and Cat ECS 200 v1.1 register lists for any specific data that is missing data missing in this document. In some cases, there are multiple occurrences of same data found in the Modbus Register (Read) lists spreadsheet which can be used as alternative Modbus registers according to the customer need and requirement. This document uses read data registers according to the order of first appearance in the Modbus Register List.

Most frequently read data such as Genset AC, Power, Engine and others can be found in the first One Hundred and Fifty (150) registers of the READ Modbus List.

Event Monitoring

There are several registers user can use to monitor events over SCADA Modbus. These registers are organized according to their context in the subsection “Cat ECS Events” of this document. Out of these, the most efficient way of event monitoring is using Alarms Group.

Examples

Following examples are presented in section 11 of this document on reading and writing the most frequently asked data:

- SCADA Data Link Access Password Write
- Engine Speed Monitor and Control
- Genset Voltage Monitor and Control
- Base Load (kW) Monitor and Control
- Cat ECS Onboard Analog Input and Output Read
- Cat ECS Onboard Analog Output Write
- Cat ECS Onboard PWM Output Read and Write
- Remote Analog Input and Output Read
- Remote Analog Output Write
- Remote RTD Input Read
- Event Logs Read

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7.1 Genset

The different generator electrical values available, depending on the generator configuration and sensing transformer configuration, are listed in the following tables.

Table 7.1: Cat ECS 100 - Electrical Values Available by Generator Configuration

Parameter	Wye (or Star L1-L2-L3-N)	3-Wire Delta (L1-L2-L3)	4-Wire Delta (L2-N-L3)	4-Wire Delta Center Tap (L1-N-L3)	2-Wire, 1-Ph (L1-L2)	2-Wire, 1-Ph (L2-L3)	3-Wire 1-Ph (L1-N-L2)	3-Wire 1-Ph (L2-N-L3)
Gen Freq	Y	Y	Y	Y	Y	Y	Y	Y
VL-L AVG	Y	Y	Y	Y	Y	Y	Y	Y
VA-B	Y	Y	Y	Y	Y	N	Y	N
VB-C	Y	Y	Y	Y	N	Y	N	Y
VC-A	Y	Y	Y	Y	N	N	N	N
VL-N AVG	Y	N	Y	Y	N	N	Y	Y
VA	Y	N	Y	Y	N	N	Y	N
VB	Y	N	Y	Y	N	N	Y	Y
VC	Y	N	Y	Y	N	N	N	Y
IAVG	Y	Y	Y	Y	Y	Y	Y	Y
IA	Y	Y	Y	Y	Y	N	Y	N
IB	Y	Y	Y	Y	Y	Y	Y	Y
IC	Y	Y	Y	Y	N	Y	N	Y
Total kW (P)	Y	Y	Y	Y	Y	Y	Y	Y
A	Y	N	Y	Y	N	N	Y	N
B	Y	N	Y	Y	N	N	Y	Y
C	Y	N	Y	Y	N	N	N	Y
Total kVAr (Q)	Y	Y	Y	Y	Y	Y	Y	Y
A	Y	N	Y	Y	N	N	Y	N
B	Y	N	Y	Y	N	N	Y	Y
C	Y	N	Y	Y	N	N	N	Y
Total kVA (S)	Y	Y	Y	Y	Y	Y	Y	Y
A	Y	N	Y	Y	N	N	Y	N
B	Y	N	Y	Y	N	N	Y	Y
C	Y	N	Y	Y	N	N	N	Y

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Table 7.2: Cat ECS 100 - Electrical Values Available by Utility Configuration (AMF Control)

Parameter	Wye (or Star L1-L2-L3-N)	3-Wire Delta (L1-L2-L3)	4-Wire Delta (L2-N-L3)	4-Wire Delta Center Tap (L1-N-L3)	2-Wire, 1-Ph (L1-L2)	2-Wire, 1-Ph (L2-L3)	3-Wire 1-Ph (L1-N-L2)	3-Wire 1-Ph (L2-N-L3)
Utility Freq	Y	Y	Y	Y	Y	Y	Y	Y
VL-L AVG	Y	Y	Y	Y	Y	Y	Y	Y
VA-B	Y	Y	Y	Y	Y	N	Y	N
VB-C	Y	Y	Y	Y	N	Y	N	Y
VC-A	Y	Y	Y	Y	N	N	N	N
VL-N AVG	Y	N	Y	Y	N	N	Y	Y
VA	Y	N	Y	Y	N	N	Y	N
VB	Y	N	Y	Y	N	N	Y	Y
VC	Y	N	Y	Y	N	N	N	Y

Table 7.3: Cat ECS 200 - Electrical Values Available by Generator Configuration (Paralleling controller only)

Parameter	Wye (or Star L1-L2-L3-N)	3-Wire Delta (L1-L2-L3)	4-Wire Delta (L2-N-L3)	4-Wire Delta Center Tap (L1-N-L3)	2-Wire, 1-Ph (L1-L2)	2-Wire, 1-Ph (L2-L3)	3-Wire 1-Ph (L1-N-L2)	3-Wire 1-Ph (L2-N-L3)
Gen Freq	Y	Y	Y	Y	Y	Y	Y	Y
VL-L AVG	Y	Y	Y	Y	Y	Y	Y	Y
VA-B	Y	Y	Y	Y	Y	N	Y	N
VB-C	Y	Y	Y	Y	N	Y	N	Y
VC-A	Y	Y	Y	Y	N	N	N	N
VL-N AVG	Y	N	Y	Y	N	N	Y	Y
VA	Y	N	Y	Y	N	N	Y	N
VB	Y	N	Y	Y	N	N	Y	Y
VC	Y	N	Y	Y	N	N	N	Y
IAVG	Y	Y	Y	Y	Y	Y	Y	Y
IA	Y	Y	Y	Y	Y	N	Y	N
IB	Y	Y	Y	Y	Y	Y	Y	Y
IC	Y	Y	Y	Y	N	Y	N	Y
Total kW (P)	Y	Y	Y	Y	Y	Y	Y	Y
A	Y	N	Y	Y	N	N	Y	N
B	Y	N	Y	Y	N	N	Y	Y
C	Y	N	Y	Y	N	N	N	Y
Total kVAr (Q)	Y	Y	Y	Y	Y	Y	Y	Y
A	Y	N	Y	Y	N	N	Y	N
B	Y	N	Y	Y	N	N	Y	Y
C	Y	N	Y	Y	N	N	N	Y
Total kVA (S)	Y	Y	Y	Y	Y	Y	Y	Y
A	Y	N	Y	Y	N	N	Y	N
B	Y	N	Y	Y	N	N	Y	Y
C	Y	N	Y	Y	N	N	N	Y

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Table 7.4: Cat ECS 200 - Electrical Values Available by Utility/Load Bus Configuration (Paralleling controller only)

Parameter	Wye (or Star L1-L2-L3-N)	3-Wire Delta (L1-L2-L3)	4-Wire Delta (L2-N-L3)	4-Wire Delta Center Tap (L1-N-L3)	2-Wire, 1-Ph (L1-L2)	2-Wire, 1-Ph (L2-L3)	3-Wire 1-Ph (L1-N-L2)	3-Wire 1-Ph (L2-N-L3)
Utility Freq	Y	Y	Y	Y	Y	Y	Y	Y
VL-L AVG	Y	Y	Y	Y	Y	Y	Y	Y
VA-B	Y	Y	Y	Y	Y	N	Y	N
VB-C	Y	Y	Y	Y	N	Y	N	Y
VC-A	Y	Y	Y	Y	N	N	N	N
VL-N AVG	Y	N	Y	Y	N	N	Y	Y
VA	Y	N	Y	Y	N	N	Y	N
VB	Y	N	Y	Y	N	N	Y	Y
VC	Y	N	Y	Y	N	N	N	Y

Table 7.5: Cat ECS 200 - Electrical Values Available by Utility Configuration AMF (Generator Set controller only)

Parameter	Wye (or Star L1-L2-L3-N)	3-Wire Delta (L1-L2-L3)	4-Wire Delta (L2-N-L3)	4-Wire Delta Center Tap (L1-N-L3)	2-Wire, 1-Ph (L1-L2)	2-Wire, 1-Ph (L2-L3)	3-Wire 1-Ph (L1-N-L2)	3-Wire 1-Ph (L2-N-L3)
Utility Freq	Y	Y	Y	Y	Y	Y	Y	Y
VL-L AVG	Y	Y	Y	Y	Y	Y	Y	Y
VA-B	Y	Y	Y	Y	Y	N	Y	N
VB-C	Y	Y	Y	Y	N	Y	N	Y
VC-A	Y	Y	Y	Y	N	N	N	N
VL-N AVG	Y	N	Y	Y	N	N	Y	Y
VA	Y	N	Y	Y	N	N	Y	N
VB	Y	N	Y	Y	N	N	Y	Y
VC	Y	N	Y	Y	N	N	N	Y

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7.1.1 Genset Rating (Active Group)

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Rated Voltage	1	316	-	214	Scaling:1 V / bit Offset: 0 Range: 0 to 64255
Rated Frequency	1	317	-	215	Scaling: 1/128 Hz / bit Offset: 0 Range: 0 to 501.9922 Hz
Rated Real Power	2	318	-	216	Scaling: 1 W / bit, 1 VA/bit, 1 VAr / bit Offset: -2000000000 Range: -2000000000 to +2211081215
Rated Apparent Power	2	320	-	218	
Rated Reactive Power	2	322	-	220	

7.1.2 Genset AC

Description TCP Port: 502	Modbus Register				Scaling, Offset, Range / Status / State / Info
	Word Count	100G	200G	200P	
Generator Average Line-Line AC RMS Voltage	1	7	-	150	Scaling: 1 V / bit Offset: 0 Range: 0 to 64255
Generator Phase A Line-Line AC RMS Voltage	1	21	-	162	
Generator Phase B Line-Line AC RMS Voltage	1	22	-	163	
Generator Phase C Line-Line AC RMS Voltage	1	23	-	164	
Generator Average Line-Neutral AC RMS Voltage	1	263	-	161	
Generator Phase A Line-Neutral AC RMS Voltage	1	27	-	165	
Generator Phase B Line-Neutral AC RMS Voltage	1	28	-	166	
Generator Phase C Line-Neutral AC RMS Voltage	1	29	-	167	
Bus Average Line-Line AC RMS Voltage	1	-	-	737	
Bus Average Line-Neutral AC RMS Voltage	1	-	-	1990	
Bus Phase A L-L AC RMS Voltage	1	-	-	1991	
Bus Phase A L-N AC RMS Voltage	1	-	-	1994	
Bus Phase B L-L AC RMS Voltage	1	-	-	1992	
Bus Phase B L-N AC RMS Voltage	1	-	-	1995	
Bus Phase C L-L AC RMS Voltage	1	-	-	1993	
Bus Phase C L-N AC RMS Voltage	1	-	-	1996	
Utility Average Line-Line AC RMS Voltage	1	88	88	226	
Generator Average Line-Line AC RMS Voltage %	1	38	38	151	Scaling: 0.0078125 % / bit Offset: -251% Range: -251 to 250.99%
Utility Average Line-Line AC RMS Voltage %	1	86	86	-	
Generator/Bus Phase Difference	1	20	20	736	Scaling: 1/128 degree / bit Offset: -200 degrees Range: -200 to 301.99deg
Generator Total Percent Current	1	-	-	153	Scaling: 0.0078125 % / bit Offset: -251% Range: -251 to 250.99%
Generator Average AC RMS Current	1	8	8	152	Scaling:1 A / bit Offset: 0 Range: 0 to 64255 A
Generator Phase A AC RMS Current	1	24	24	168	
Generator Phase B AC RMS Current	1	25	25	169	
Generator Phase C AC RMS Current	1	26	26	170	
Generator Phase A Frequency	1	273	-	171	Scaling: 1/500 Hz / bit Offset: 0 Range: 0 to 131.072
Generator Phase B Frequency	1	274	-	172	
Generator Phase C Frequency	1	275	-	173	

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Genset AC - continued

Description TCP Port: 502	Modbus Register				Scaling, Offset, Range / Status / State / Info
	Word Count	100G	200G	200P	
Generator Average AC RMS Frequency Percent	1	39	39	158	Scaling: 0.0078125 % / bit Offset: -251% Range: -251 to 250.99%
Generator Average AC RMS Frequency	1	9	9	157	Scaling: 1/128 Hz / bit Offset: 0 Range: 0 to 501.9922 Hz
Bus Average AC RMS Frequency	1	19	19	735	
Utility Average AC RMS Frequency	1	87	87	227	
Generator Overall Power Factor	1	10	10	159	Scaling: 1/16384 / bit Offset: -1 Range: -1.0 to 1.0
Generator Phase A Power Factor	1	288	-	186	
Generator Phase B Power Factor	1	289	-	187	
Generator Phase C Power Factor	1	290	-	188	
Generator Desired Power Factor	1	1449	-	647	
Generator Overall Power Factor Lagging	1	11	11	160	0 = leading PF 1 = lagging PF
Generator Phase A Power Factor Lagging	1	291	-	189	
Generator Phase B Power Factor Lagging	1	292	-	190	
Generator Phase C Power Factor Lagging	1	293	-	191	

7.1.3 Genset Power

0	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Generator Total Real Power	2	12	12	155	Scaling: 1 W / bit, 1 VAr/bit, 1 VA / bit Offset: -2000000000 Range: -2000000000 to +2211081215
Generator Total Reactive Power	2	14	14	205	
Generator Total Apparent Power	2	16	16	202	
Generator Phase A Real Power	2	276	-	174	
Generator Phase B Real Power	2	278	-	176	
Generator Phase C Real Power	2	280	-	178	
Generator Phase A Reactive Power	2	282	-	180	
Generator Phase B Reactive Power	2	284	-	182	
Generator Phase C Reactive Power	2	286	-	184	
Generator Total Real Power Percent	1	36	-	154	Scaling: 0.0078125 % / bit Offset: -251% Range: -251 to 250.99%
Generator Total Reactive Power Percent	1	37	-	207	
Generator Total Apparent Power Percent	1	306	-	204	

7.1.4 Genset Energy

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Generator Total Real Power Energy in kWh	2	30	-	208	Scaling: 1 kWh / bit, 1kVAr / bit Offset: 0 Range: 0 to 4211081215
Generator Total Reactive Power Export in kVArh	2	32	-	210	
Generator Total Reactive Power Import in kVArh	2	34	-	212	

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7.1.5 State, Mode and Status

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Engine Operating Mode	1	1	1	-	0 = Stop 1 = Auto 2 = Run
Automatic Start/Stop State	1	6	6	-	0 = Init 1 = Pre Crank 2 = Starting 3 = Running 4 = Pre Cooldown 5 = Cooldown 6 = Stopping 7 = Stopped 8 = Idling
Sync Mode Switch State	1	-	-	738	0 = Off 1 = Automatic 2 = Manual 3 = Check
Unit Dead Bus Arbitration State	1	-	-	730	1 = DBA Disabled 2 = DBA Enabled 3 = Requesting 4 = Capturing 5 = Close Granted 6 = Breaker Closed 7 = Failure Wait 8 = Live Bus
System Dead Bus Arbitration state	1	-	-	731	1 = DBA Disabled 2 = DBA Enabled 3 = Requesting 4 = Capturing 5 = Close Granted 6 = Breaker Closed 7 = Failure Wait 8 = Live Bus
Generator Circuit Breaker Status	1	153	96	229	Status Indicated By First Three Bits Of The 16-Bit Register Word Bit (3|2|1) 000 = Open 001 = Closed 010 = Locked Out 011 = Not Available 100 = Not Installed
Utility Circuit Breaker Status	1	1707	97	225	
BUS live status	1	-	-	228	Bit 0 : Gen Bus Bit 1 : Load Bus
Live Bus Indication	1	-	-	230	Bit 0 : Gen Bus Bit 1: Load Bus Bit 2: Utility Status

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7.1.6 Genset Start / Stop

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Cooldown Duration Remaining	1	95	95	-	Scaling: 1 second / bit Offset: 0 Range: 0 to 64255 seconds
Number of Crank Attempts	2	89	89	-	Scaling: 1 second / bit Offset: 0 Range: 0 to 4211081215
Number of Successful Starts	2	91	91	-	
Bypass Cooldown Command	1	107	101	-	0 = False 1 = True
Description (Commands) TCP Port: 50200	**Word Count**	**Modbus Register**			**Scaling, Offset, Range / Status / State / Info**
		100G	**200G**	**200P**	
Engine Control Switch Command	1	108	102	-	0 = Stop 1 = Auto 2 = Run
Group Start Command	1	-	-	103	00 = False 01 = True 10 = Error 11 = Disabled Or Not Available
Remote Initiate Command *when Cat ECS is in Auto position, sending True commands the engine to start.*	1	18	103	-	
Sync Mode Switch Command	1	-	-	104	0 = Off 1 = Automatic 2 = Manual 3 = Check
Genset CB Close Inhibit	1	-	-	7	00 = FALSE 01 = TRUE 10 = ERROR 11 = DISABLED OR NOT AVAILABLE

7.1.7 Battery Voltage

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Battery Voltage	1	40	40	-	Scaling: 0.05 V / bit Offset: 0 Range: 0 to 3212.75 V

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7.1.8 Base Load, Power Factor and Reactive Load (Write Commands)

Description TCP Port: 50200	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Generator Desired Power Factor Command (Read & Write)	1	-	-	8	Scaling: 1/16384 / bit Offset: -1.0 Range: -1.0 to 1.0
Desired Base Load Command (Read & Write)	2	-	-	9	Scaling: 1 W / bit Offset: -2000000000 Range: -2000000000 to +2211081215
Reactive Load Control Reactive Power Setting (Read & Write)	2	-	-	14	Scaling: 1 VAr / bit Offset: -2000000000 Range: -2000000000 to +2211081215
Reactive Load Control Mode Configuration	1	-	-	13	0= Constant Reactive Power 1= Constant Power Factor
Reactive Load Control Non-Grid Code Mode Reactive Power Source Configuration	1	-	-	16	0 = Configured Value 1 = Analog 2 = Modbus 3 = Programmable Logic Function
Reset Load Shed	1	-	-	6	00 = FALSE 01 = TRUE 10 = ERROR 11 = DISABLED OR NOT AVAILABLE

7.1.9 Load Sense Load Demand

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
APCDL Expected Units Online	2	-	-	752	bit1=Power System Unit Number 1 bit2=Power System Unit Number 2 . . Bit32=Power System Unit Number 32
APCDL Actual Units Online	2	-	-	754	
APC DL Unit Load Sense Load Demand State	1	-	-	1466	0 = GOING 1 = SETTLING 2 = UNLOADING 3 = LOADING 4 = AVAILABLE 5 = GOING LSLD OFF 6 = UNAVAILABLE
APC DL System Load Sense Load Demand State	1	-	-	1467	
APC DL System Load Sense Load Demand Sequence Strategy	1	-	-	1470	

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Load Sense Load Demand - continued

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
System LSLD Unit Sequence Priority Number	1	-	-	1471	Scaling: 1 / bit Offset: 0 Range: 0 to 65535
APCDL Expected Number of Units Online	1	-	-	750	
APCDL Actual Number Of Units Online	1	-	-	751	
Load Sense Load Demand Proactive Add Units	1	-	-	1465	
Load Sense Load Demand Number of Redundant Units	1	-	-	1472	
Active Rating Group	1	-	-	1477	
APC DL Unit Load Sense Load Demand Load/Unload Sequence # *(use in conjunction with register 1468).*	1	-	-	1469	
APC DL Unit Load Sense Load Demand Load/Unload Sequence *(use in conjunction with register 1469).*	1	-	-	1468	0 = NEXT TO LOAD 1 = NEXT TO UNLOAD
LSLD Real Power Spinning Reserve	2	-	-	1473	Scaling: 1 W / bit Offset: -2000000000 Range: '-2000000000 to +2211081215 W
Actual System Real Power kW	2	-	-	756	Scaling: 1 W / bit Offset: -2000000000 Range: '-2000000000 to +2211081215 W
APCDL System Total Real Power Percent	1	-	-	758	Scaling: 0.0078125 % / bit Offset: -251 % Range: -251 to 250.99 %
Actual System Reactive Power Percent	1	-	-	1475	
Actual System Real Power Percent	1	-	-	1476	

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7.1.10 Load Share and Load Control

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Generator Real Load Control State	1	-	-	1805	0 = LOAD SHARE - DISABLED 1 = LOAD SHARE - INACTIVE 2 = LOAD SHARE - SOFT LOAD 3 = LOAD CONTROL - SOFT LOAD 4 = LOAD SHARE - SOFT UNLOAD 5 = LOAD CONTROL - SOFT UNLOAD 6 = LOAD SHARE - FAILSAFE DROOP 7 = LOAD SHARE - FAILSAFE ISOCH 8 = LOAD CONTROL - BASE LOAD 9 = LOAD CONTROL - PF CONTROL 11 = LOAD SHARE - MGDL/APCDL
Unit Load Sharing State #1	1	-	-	764	
Generator Desired kVAr	2	1445	-	645	Scaling: 1 VAr / bit Offset: -2000000000 Range: -2000000000 to +2211081215 Var
Generator Desired Power Factor	1	1449	-	647	Scaling: 1/16384 / bit Offset: -1 Range: -1.0 to 1.0
Generator Real Load Control Base Load Setpoint	1	-	-	1737	Scaling: 0.01% / bit Offset: -321.27 Range: -321.27% to 321.28%
Generator Total Real Power	2	12	12	155	Scaling: 1 W / bit, 1 VAr/bit, 1 VA / bit Offset: -2000000000 Range: -2000000000 to +2211081215
Generator Total Reactive Power	2	14	14	205	
Generator Total Apparent Power	2	16	16	202	
Generator Total Real Power Percent	1	36	36	154	Scaling: 0.0078125 % / bit Offset: -251% Range: -251 to 250.99%
Generator Total Reactive Power Percent	1	37	37	207	
Generator Total Percent Apparent Power Percent	1	306	-	204	

7.1.11 Utility Transfer (Write Command)

Description TCP Port: 50200	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Bypass Transfer Delay Command	1	109	-	11	00 = FALSE 01 = TRUE
Transfer to Generator Power Command	1	110	-	12	00=Don't care. 01=Utility 10= Gen

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7.1.12 Generator Winding and Bearing Temperature

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Generator Phase A Winding Temperature from Data Link	1	297	162	-	Scaling: 0.03125 C / bit Offset: -273 Range: -273 to 1735 C
Generator Phase B Winding Temperature from Data Link	1	298	163	-	
Generator Phase C Winding Temperature from Data Link	1	299	164	-	
Generator Front Bearing Temperature from Data Link (bearing #2 or right bearing)	1	300	165	-	
Generator Rear Bearing Temperature from Data Link (bearing #1 or left bearing)	1	301	166	-	Scaling: 0.03125 C / bit Offset: -273 Range: -273 to 1735 C

7.1.13 Gas Substitution (Write Command)

Description TCP Port: 50200	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Maximum Gas Substitution Percentage Increase Command	1	8	3	-	Scaling: 0.0078125 % / bit Offset: -251 Range: -251 to 250.99 %
Requested Gas Substitution Mode	1	9	4	-	Scaling: 1 / bit Offset: 0 Range: 0 to 1

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7.1.14 Generator Paralleling Data (Gen #1 to Gen #32)

This data set from Gen #1 to Gen #n (32) is available on each Cat ECS 200 Paralleling Genset controller and can be used as an alternative path to read multiple genset data listed below via single or multiple Cat ECS 200 paralleling controllers.

Description TCP Port: 502	Word Count	Modbus Register 200P	Scaling, Offset, Range / Status / State / Info
Warning Unit #1	1	759	00 = False 01 = True 10 = Error 11 = Disabled Or Not Available
Shutdown unit #1	1	760	
APC DL Load Sense Load Demand State Unit #1	1	761	0 = Going 1 = Settling 2 = Unloading 3 = Loading 4 = Available 5 = Going LSLD Off 6 = Unavailable
LSLD Unit Sequence Priority Number Unit #1	1	762	Scaling: 1 / bit Offset: 0 Range: 0 to 65535
Dead Bus Arbitration State Unit #1	1	763	1 = DBA Disabled 2 = DBA Enabled 3 = Requesting 4 = Capturing 5 = Close Granted 6 = Breaker Closed 7 = Failure Wait 8 = Live Bus
Unit Load Sharing State #1	1	764	0 = Load Share - Disabled 1 = Load Share - Inactive 2 = Load Share - Soft Load 3 = Load Control - Soft Load 4 = Load Share - Soft Unload 5 = Load Control - Soft Unload 6 = Load Share - Failsafe Droop 7 = Load Share - Failsafe Isoch 8 = Load Control - Base Load 9 = Load Control - PF Control 11 = Load Share - MGDL
Engine Operating Hours Unit #1	2	765	Scaling: 0.05 hr / bit Offset: 0 hr Range: 0 to 210554060.75 hr
Engine Operating Mode Unit #1	1	767	0 = Stop 1 = Auto 2 = Run
Sync Mode Switch State #1	1	768	0 = Off 1 = Automatic 2 = Manual 3 = Check
Generator Total Reactive Power Unit #1	2	769	Scaling: 1 VAr / bit Offset: - 2000000000 VAr Range: -2000000000 to +2211081215

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Generator Paralleling Data (Gen #1 to Gen #32) - continued

Description TCP Port: 502	Word Count	Modbus Register 200P	Scaling, Offset, Range / Status / State / Info
Generator Total Real Power Unit #1	2	771	Scaling: 1 W / bit Offset: - 2000000000 W Range: -2000000000 to +2211081215
Generator Total Percent kVAr Unit #1	1	773	Scaling: 0.0078125 % / bit Offset: -250 % Range: -250 to 250.99 %
Generator Total Percent kW Unit #1	1	774	
Auto Start Stop State unit #1	1	775	0 = INIT 1 = PRE_CRANK 2 = STARTING 3 = RUNNING 4 = PRE_COOL 5 = COOLDOWN 6 = STOPPING 7 = STOPPED 8 = IDLING
Generator Average Line-Line AC RMS Voltage Unit #1	1	776	Scaling: 1 V / bit Offset: 0 V Range: 0 to 64255 V
Generator Average AC RMS Frequency Unit #1	1	777	Scaling: 1/128 Hz / bit Offset: 0 Hz Range: 0 to 501.9922 Hz
Generator Breaker Status Unit #1	1	778	000 = Open 001 = Closed 010 = Locked Out 011 = Not Available 100 = Not Installed
Bus Average Line-Line AC RMS Voltage Unit #1	1	779	Scaling: 1 V / bit Offset: 0 V Range: 0 to 64255 V
Live Bus Indication unit #1	1	780	Bit 0 : Gen Bus Bit 1: Load Bus Bit 2: Utility Status
Generator Real Load Control State	1	1805	0 = Load Share - Disabled 1 = Load Share - Inactive 2 = Load Share - Soft Load 3 = Load Control - Soft Load 4 = Load Share - Soft Unload 5 = Load Control - Soft Unload 6 = Load Share - Failsafe Droop 7 = Load Share - Failsafe Isoch 8 = Load Control - Base Load 9 = Load Control - PF Control 10 = Load Share - Hard-Wired 11 = Load Share - MGDL

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7.2 Engine

7.2.1 Service Meter

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Engine Operating Hours	2	43	154	-	Scaling: 0.05 hour / bit Offset: 0 Range: 0 to 210554060.75 hour
Service Maintenance Interval Hours Remaining	1	419	211	-	Scaling: 0.05 hour / bit Offset: -32127 Range: -32127 to 32128 hour
Service Maintenance Interval days Remaining	1	420	212	-	Scaling: 0.05 day / bit Offset: -32127 Range: -32127 to 32128 days

7.2.2 Engine Speed

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Engine Speed	1	165	159	-	Scaling: 0.125 rpm / bit Offset: 0 Range: 0 to 8031.875 rpm

7.2.3 Engine Fuel

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Fuel Level from Data Link	1	55	55	-	Scaling: 0.0078125 % / bit Offset: -251 Range: -251 to 250.99 %
Fuel Level - Analog *takes precedence over datalink.*	1	151	151	-	
Fuel Level *As received from I/O pin analog input (recommend using 151 instead)*	1	462	253	-	
Engine Fuel Temperature from Data Link	1	60	60	-	Scaling: 0.03125 C / bit Offset: -273 Range: -273 to 1735 C
Fuel Pressure from CAN Data Link	1	61	61	-	Scaling: 0.125 kPa / bit Offset: 0 Range: 0 to 8031.875 kPa
Total Fuel Consumption from Data Link	2	47	47	-	Scaling: 0.5 L / bit Offset: 0 Range: 0 to 2105540607.5 L
Fuel Load Pump Active Status	1	1300	940	-	0 = False 1 = True 2 = Error 3 = Disabled Or Not Available
Fuel Pump Primer Status	1	1630	1050	-	00 = Priming Inactive 01 = Priming Active 10 = Priming Inappropriate 11 = Data Unavailable

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Engine Fuel - continued

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Manual Fuel Priming Duration Remaining	1	1631	1051	-	Scaling: 0.25 seconds / bit Offset: 0 Range: 0 to 62.5 seconds
Fuel Shutoff 1 Control from Data Link	1	492	282	-	0 = Open 1 = Closed 2 = Locked Out 7 = Error
Fuel Transfer - Fuel Load Pump Active	1	1300	940	-	0 = False 1 = True 2 = Error 3 = Disabled Or Not Available

7.2.4 Engine Oil

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Engine Oil Pressure	1	41	41	-	Scaling: 0.125 kPa / bit Offset: 0 Range: 0 to 8031.875 kPa
Oil Filter Differential Pressure from Data Link	1	53	53	-	
Engine Oil Temperature	1	59	59	-	Scaling: 0.03125 C / bit Offset: -273 Range: -273 to 1735 C
Engine Pre-filter Oil Pressure from Data Link	1	85	85	-	Scaling: 4 kPa / bit Offset: 0 Range: 0 to 1000 kPa
Engine Oil Priming State	1	536	326	-	0 = False 1 = True 2 = Error 3 = Disabled Or Not Available
Engine Oil Priming Switch	1	537	327	-	

7.2.5 Engine Coolant

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Engine Coolant Temperature	1	42	42	-	Scaling: 0.03125 C / bit Offset: -273 Range: -273 to 1735 C
Engine Oil to Engine Coolant Differential Temperature	1	427	219	-	Scaling: 1.0 C / bit Offset: -273 Range: -273 to 32767 C
Engine Coolant Temperature 2	1	428	220	-	Scaling: 1 C / bit Offset: -40 Range: -40 to 210 C
Engine Coolant Temperature 3	1	429	221	-	
Engine Coolant Pressure from Data Link	1	440	231	-	Scaling: 0.125 kPa / bit Offset: 0 Range: 0 to 8031.875 kPa

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7.2.6 Engine Exhaust

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Exhaust Manifold #1 (Left) Temperature from Data Link	1	57	57	-	Scaling: 0.03125 C / bit Offset: -273 Range: -273 to 1735 C
Exhaust Manifold #2 (Right) Temperature from Data Link	1	58	58	-	
Exhaust Gas Temperature Average from Data Link	1	670	354	-	
Exhaust Temperature from Data Link	1	676	357	-	
Cylinder #1 - 20 Exhaust Port Temperature from Data Link	20	650 - 669	334-353	-	
Cylinder #21 Exhaust Port Temperature from Data Link	1	777	456	-	
Cylinder #22 Exhaust Port Temperature from Data Link	1	782	461	-	
Cylinder #23 Exhaust Port Temperature from Data Link	1	787	466	-	
Cylinder #24 Exhaust Port Temperature from Data Link	1	792	471	-	

7.2.7 Engine Intake

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Intake Manifold #1 Temperature from Data Link	1	46	46	-	Scaling: 0.03125 C / bit Offset: -273 Range: -273 to 1735 C

7.2.8 Engine Turbocharger

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Turbocharger 1-4 Compressor Inlet Temperature from CAN Data Link	4	49 - 52	49-52	-	Scaling: 0.03125 C / bit Offset: -273 Range: -273 to 1735 C
Turbocharger 1 Turbine Inlet Temperature from CAN Data Link	1	67	67	-	
Turbocharger 1 Turbine Outlet Temperature from CAN Data Link	1	68	68	-	Scaling: 0.03125 C / bit Offset: -273 Range: -273 to 1735 C
Turbocharger 2 Turbine Inlet Temperature from CAN Data Link	1	69	69	-	
Turbocharger 2 Turbine Outlet Temperature from CAN Data Link	1	70	70	-	
Turbocharger 3 Turbine Inlet Temperature from CAN Data Link	1	71	71	-	
Turbocharger 3 Turbine Outlet Temperature from CAN Data Link	1	72	72	-	
Turbocharger 4 Turbine Inlet Temperature from CAN Data Link	1	73	73	-	
Turbocharger 4 Turbine Outlet Temperature from CAN Data Link	1	74	74	-	

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Engine Turbocharger - continued

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Boost Pressure from CAN Data Link	1	63	63	-	Scaling: 0.125 kPa / bit Offset: 0 Range: 0 to 8031.875 kPa
Left Turbo Inlet Pressure from CAN Data Link	1	76	76	-	Scaling: 1 / 128 kPa / bit Offset: -250 Range: -250 to 251.99 kPa
Right Turbo Inlet Pressure from CAN Data Link	1	77	77	-	
Engine Turbocharger Compressor Bypass Actuator 1 Position	1	80	80	-	Scaling: 0.4 % / bit Offset: 0 Range: 0 to 100%

7.2.9 Engine Filter

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Air Filter 1 Differential Pressure from Data Link	1	45	45	-	Scaling: 1 / 128 kPa / bit Offset: -250 Range: -250 to 251.99 kPa
Oil Filter Differential Pressure from Data Link	1	53	53	-	Scaling: 0.125 kPa / bit Offset: 0 Range: 0 to 8031.875 kPa
Fuel Filter Differential Pressure from Data Link	1	54	54	-	
Engine Air Filter #2 Differential Pressure	1	64	64	-	Scaling: 0.05 kPa / bit Offset: 0 Range: 0 to 12.5 kPa
Engine Pre-filter Oil Pressure from Data Link	1	85	85	-	Scaling: 4 kPa / bit Offset: 0 Range: 0 to 12.5 kPa

7.2.10 Engine Aftertreatment

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Urea Tank Level from CAN Data Link	1	472	263	-	Scaling: 0.4 % / bit Offset: 0 Range: 0 to 100%
Urea Injection Air Supply Pressure from CAN Data Link	1	474	265	-	Scaling: 0.1 kPa / bit Offset: 0 Range: 0 to 6425.5 kPa
Aftertreatment #1 SCR Catalyst Reagent Tank #1 Temperature from CAN Data Link	1	473	264	-	Scaling: 1 C / bit Offset: -40 Range: -40 to 210 C
Catalyst Intake Temperature	1	475	266	-	Scaling: 0.03125 C / bit Offset: -273 Range: -273 to 1735 C

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Engine Aftertreatment - continued

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Aftertreatment 1 SCR Catalyst Exhaust Gas Differential Pressure from CAN Data Link	1	476	267	-	Scaling: 0.1 kPa / bit Offset: 0 Range: 0 to 6425.5 kPa
Aftertreatment Soot Load Level	1	477	268	-	Scaling: 1 % / bit Offset: 0 Range: 0 to 250 %
SCR System State	1	478	269	-	0000 ($0) = Sleep Mode 0001 ($1) = Preparing 0010 ($2) = Normal 0011 ($3) = System Error Pending 0100 ($4) = DEF Purging 0101 ($5) = Heat Protection 0110 ($6) = Cold Protection 0111 ($7) = Shutoff 1000 ($8) = Diagnosis 1001 ($9) = Service Mode, Dose Allowed 1010 ($10) = Service Mode, Dose Not Allowed / Dose Inhibit 1011 ($11) = Ok to Power Down 1100 ($12) = Reserved for future use. 1101 ($13) = Reserved for future use. 1110 ($14) = Error 1111 ($15) = Not Available
Aftertreatment DEF Dosing Temperature from CAN	1	479	270	-	Scaling: 1 C / bit Offset: -40 Range: -40 to 210 C
Aftertreatment Total DEF Used from CAN	2	480	271	-	Scaling: 0.5 L / bit Offset: 0 Range: 0 to 2105540607.5 L
Time Since Aftertreatment Inducement from CAN	1	482	273	-	Scaling: 0.2 hour / bit Offset: 0 Range: 0 to 12851 hours
Time Until Aftertreatment Inducement from CAN	1	483	274	-	Scaling: 0.25 hour / bit Offset: 0 Range: 0 to 62.5 hours
SCR Operator Inducement Override Mode	1	484	275	-	000 = Not Enabled 001 = Available Not Active 010 = Active 011 = Active Extended 100 = Expired 101 = Not Defined 110 = Not Defined 111 = Not Available
Three Way Catalyst Intake Temperature from CAN Data Link	1	507	297	-	Scaling: 0.03125 C / bit Offset: -273 Range: -273 to 1735 C

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7.2.11 Engine Gas

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Cylinder #1 - 20 Knock Level	20	736 - 755	416-435	-	Scaling: 0.5 kPa / bit Offset: 0 Range: 0 to 125 kPa
Desired Ignition Timing	1	500	290	-	Scaling: 1 / 128 degree / bit Offset: -200 Range: -200 to 301.99 degrees
Engine Actual Ignition Timing	1	501	291	-	
Cylinder # 1 - 20 Ignition Timing	20	757 - 776	436-455	-	
Cylinder #1 - 20 Ignition Transformer Secondary Output from Data Link	20	693 - 712	373-392	-	Scaling: 1% / bit Offset: -1.25 Range: -125% to 125%
Cylinder #21 Ignition Transformer Secondary Output from Data Link	1	780	459	-	
Cylinder #22 Ignition Transformer Secondary Output from Data Link	1	785	464	-	
Cylinder #23 Ignition Transformer Secondary Output from Data Link	1	790	469	-	
Cylinder #24 Ignition Transformer Secondary Output from Data Link	1	795	474	-	Scaling: 1% / bit Offset: -1.25 Range: -125% to 125%
Specific Heat Ratio	1	466	257	-	Scaling: 0.001 / bit Offset: 0 Range: 0 to 64.255
Engine Fuel Specific Gravity	1	465	256	-	Scaling: 0.0001 / bit Offset: 0 Range: 0 to 6.4255
Engine Inlet Air Mass Flow Rate	1	468	259	-	Scaling: 0.05 kg/h per bit Offset: 0 Range: 0 to 3212.75 kg/h
Engine Gaseous Fuel Correction Factor	1	464	255	-	Scaling: 0.5 kPa / bit Offset: 0 Range: 0 to 125 kPa
Engine Fuel Valve Differential Pressure	1	446	237	-	Scaling: 0.1 kPa / bit Offset: 0 Range: 0 to 6425.5 kPa
Fuel Valve 1 Inlet Absolute Pressure from Data Link	1	469	260	-	Scaling: 0.1 kPa / bit Offset: 0 Range: 0 to 6425.5 kPa
Requested Engine Fuel Valve #1 Position	1	503	293	-	Scaling: 0.4 % / bit Offset: 0 Range: 0 to 100%
Engine Throttle Valve Position	1	504	294	-	
Desired Engine Throttle Valve Position	1	505	295	-	
Oxygen Sensor Status	1	512	302	-	Scaling: 1 / bit Offset: 0 Range: 0 to 1

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Engine Gas - continued

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Oxygen Sensor Closed Loop Operation	1	513	303	-	00 = Open Loop 01 = Closed Loop 10 = Reserved 11 = Don't Care / Take No Action
Intake Oxygen Sensor Voltage	1	514	304	-	Scaling: 0.5 mV / bit Offset: 0 Range: 0 to 32127.5 mV
Desired Intake Oxygen Sensor Voltage	1	515	305	-	
Outlet Oxygen Sensor Voltage	1	516	306	-	
Desired Outlet Oxygen Sensor Voltage	1	517	307	-	
Intake Oxygen Sensor Fuel Trim Percentage	1	518	308	-	Scaling: 0.01 % / bit Offset: -320 Range: -320 to 322.55 %
Outlet Oxygen Sensor Fuel Trim Percentage	1	519	309	-	
Exhaust NOx Level	1	529	319	-	Scaling: 0.05 ppm / bit Offset: 0 Range: 0 to 3275.15 ppm
Engine Starter Mode	1	539	329	-	0000 = Start Not Requested (Not Req) 0001 = Starter Active, Gear Not Engaged (Not Engage) 0010 = Starter Active, Gear Engaged (Engaged) 0011 = Start Finished, Starter Not Active (Not Active) 0100 = Starter Inhibited, Engine Running (Running) 0101 = Starter Inhibited, Engine Not Ready (Not Ready) 0110 = Starter Inhibited, Driveline Engaged (Driveline) 0111 = Starter Inhibited, Active Immobilizer (Immobilizer) 1000 = Starter Inhibited, Over-Temp (Over Temp) 1001 = Starter Inhibited, Air Shutoff Valve Active (Air Shutoff) 1010 = Starter Inhibited, SCR Inducement Active (SCR Active) 1011 = Reserved (Reserved) 1100 = Starter Inhibited, Reason Unknown (Unknown) 1101 = Error (Legacy) (Error) 1110 = Error (Error) 1111 = Not Available (Not Avail)
Desired Exhaust NOx Level	1	530	320	-	Scaling: 0.05 ppm / bit Offset: 0 Range: 0 to 3275.15 ppm
Exhaust NOx Level Sensor Status	1	531	321	-	0 =Off 14 = Active 15 = Inactive 1004 = Heating

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Engine Gas - continued

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Fuel Energy Content	1	463	254	-	Scaling: 1 / 256 MJ / Nm^3 per bit Offset: 0 Range: 0 to 250.996 MJ / Nm^3
Desired Air to Fuel Ratio	1	523	313	-	Scaling: 1 / bit Offset: 0 Range: 0 to 65.503 11 = Disabled Or Not Available
Actual Air to Fuel Ratio	1	524	314	-	Scaling: 0.1 / bit Offset: 0 Range: 0.0 to 30.0

7.2.12 Engine Dynamic Gas Blending

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Gas Substitution System Enable Status	1	491	1216	-	00 = Disabled 01 = Enabled
Gas Substitution Status	1	487	278	-	00 = Inactive 01 = Active 10 = Locked Inactive
Maximum Gas Substitution Percentage	1	488	279	-	Scaling: 0.0078125 % / bit Offset: -251 Range: 0 - 100%
Actual Gas Substitution Percentage	1	490	281	-	

7.2.13 Engine – Other Data

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Instantaneous Fuel Consumption from Data Link	1	65	65	-	Scaling: 0.05 L/h per bit Offset: 0 Range: 0 to 3212.75 L/h
Atmospheric Pressure from Data Link	1	66	66	-	Scaling: 0.125 kPa / bit Offset: 0 Range: 0 to 8031.875 kPa
Crankcase Pressure from Data Link	1	62	62	-	Scaling: 1 / 128 kPa / bit Offset: -250 Range: -250 to 251.99 kPa
Engine Ventilation Status	1	521	311	-	00 = FALSE 01 = TRUE

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7.3 Programmable Cycle Timer

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Programmable Cycle Timer 1 Output 1 Status	1	1601	1053	-	bits 15:2 UNUSED bits 1:0 Timer Status
Programmable Cycle Timer 2 Output 1 Status	1	1602	1054	-	
Programmable Cycle Timer 3 Output 1 Status	1	1603	1055	-	
Programmable Cycle Timer 4 Output 1 Status	1	1604	1056	-	
Programmable Cycle Timer 5 Output 1 Status	1	1605	1057	-	
Programmable Cycle Timer 6 Output 1 Status	1	1606	1058	-	
Programmable Cycle Timer 7 Output 1 Status	1	1607	1059	-	
Programmable Cycle Timer 1 Output 2 Status	1	1608	1060	-	
Programmable Cycle Timer 2 Output 2 Status	1	1609	1061	-	
Programmable Cycle Timer 3 Output 2 Status	1	1610	1062	-	
Programmable Cycle Timer 4 Output 2 Status	1	1611	1063	-	
Programmable Cycle Timer 5 Output 2 Status	1	1612	1064	-	
Programmable Cycle Timer 6 Output 2 Status	1	1613	1065	-	
Programmable Cycle Timer 7 Output 2 Status	1	1614	1066	-	
Programmable Cycle Timer 1 Output 3 Status	1	-	1067	-	
Programmable Cycle Timer 2 Output 3 Status	1	-	1068	-	
Programmable Cycle Timer 3 Output 3 Status	1	-	1069	-	
Programmable Cycle Timer 4 Output 3 Status	1	-	1070	-	
Programmable Cycle Timer 5 Output 3 Status	1	-	1071	-	
Programmable Cycle Timer 6 Output 3 Status	1	-	1072	-	
Programmable Cycle Timer 7 Output 3 Status	1	-	1073	-	

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7.4 Realtime Clock

Description	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
TCP Port: 502 Real Time Clock (Read)	3	1690	-	245	Byte 5 = J1939 Time 14 (Year) Byte 4 = J1939 Time 11 (Quarter Day) Byte 3 = J1939 Time 13 (Month) Byte 2 = J1939 Time 7 (Hour) Byte 1 = J1939 Time 6 (Minute) Byte 0 = J1939 Time 3 (Quarter Second)
TCP Port: 50200 Real Time Clock Update Command *SCADA command to change the real time clock information*	3	5	-	3	Byte 4 = J1939 Time 11 (Quarter Day) Byte 3 = J1939 Time 13 (Month) Byte 2 = J1939 Time 7 (Hour) Byte 1 = J1939 Time 6 (Minute) Byte 0 = J1939 Time 3 (Quarter Second)

7.4.1 Realtime Clock Data Consist

Register (100G)	Register (200P)	Register (100G)	Register (200P)	Byte	MSB / LSB	Scaling	Format	Range
Read		Write						
1690	245	5	3	1	MSB	0.25 seconds / bit	0 seconds (HH:MM:00 = 0x00)	0 to 62.5 seconds
				0	LSB	1 minute / bit	0 minutes (HH:00 = 0x00)	0 to 250 minutes
1691	246	6	4	1	MSB	1 hour / bit	0 hours (midnight = 0x00)	0 to 250 hours
				0	LSB	1 month / bit	1 month (Jan = 0x01)	0 to 250 months
1692	247	7	5	1	MSB	0.25 days / bit	0.75 days (12:00-6:00am on day 1 = 0x01)	0 to 62.5 days
				0	LSB	1 year / bit	1985 years	1985 to 2235 years

Note: See sections “Byte Extracting” and “Byte Packing” on how to read and extract the bytes from the word and how to pack them for Cat ECS writes.

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7.5 Cat ECS Inputs and Outputs

The Cat ECS controllers support following number of Onboard and Remote Inputs and Outputs:

Onboard Inputs and Outputs	Remote Inputs and Outputs
• Digital Inputs: 16; Isolated Digital Input: 1 • Digital Outputs: 12 • Analog Inputs: 3 • Analog Outputs: 2 • PWM Outputs: 3	• Remote Digital Inputs: 48 • Remote Digital Outputs: 48 • Remote Analog Inputs: 8 • Remote Analog Outputs: 8 • Remote RTD Inputs: 4

7.5.1 Read - Onboard Inputs and Outputs

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Digital Input #1 to Input# 16 Bit-0 to Bit-15: DI-1 to DI-16	1	1640	1080	700	0 = Deactivated 1 = Activated
Bit-0: IDI-1 (Isolated Digital Input) Bit-1: Digital Output-1 Bit-12: Digital Output-12 *Bit 13-15: Not Used*	1	1641	1081	701	
Analog Input #1 Sensor Value	1	1642	1082	702	Scaling: 1/16 / bit Offset: -16 Range: -16 to 3999.9375 *Raw sensor value as read from the input (Vdc)*
Analog Input #2 Sensor Value	1	1643	1083	703	
Analog Input #3 Sensor Value	1	1644	1084	704	
Analog Input #1 Data Value	2	1645	1085	705	Scaling: 1/256 / bit Offset: -8,224,768 Range: -8,224,768 to 8,224,767.99609375 *Data value as read from the input, in engineering units.*
Analog Input #2 Data Value	2	1647	1087	707	
Analog Input #3 Data Value	2	1649	1089	709	
Analog Output #1 Data Value	2	1651	1091	711	
Analog Output #2 Data Value	2	1653	1093	713	
Analog Output #1 Output Value	1	1655	1095	715	Scaling: 1/16 / bit Offset: -16 Range: -16 to 3999.9375 *Indicates the physical value for Analog output when configured for Type Current*
Analog Output #2 Output Value	1	1656	1096	716	
PWM #1 Data Value	2	1657	1097	717	Scaling: 1/256 / bit Offset: -8,224,768 Range: -8,224,768 to 8,224,767.99609375 *Value to be output via PWM output, in engineering units.*
PWM #2 Data Value	2	1659	1099	719	
PWM #3 Data Value	2	1661	1101	721	
PWM #1 Output Value	1	1663	1103	723	Scaling: 0.0025 % / bit Offset: 0 Range: 0 to 160.6375 % *Indicates the output percentage for PWM Output*
PWM #2 Output Value	1	1664	1104	724	
PWM #3 Output Value	1	1665	1105	725	

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7.5.2 Write - Onboard Outputs

Description TCP Port: 50200	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Digital Output #1 to Digital Output # 12 *Digital Output Command per Word x Total of 12 Words*	1	21 to 32	25 to 36	27 to 38	0 = Deactivate 1 = Activate *Command to activate this output. Applies only when Usage Type is set to Data Link.*
Analog Output #1	2	33	15	17	Scaling: 1/256 / bit Offset: -8,224,768 Range: -8,224,768 to 8,224,767.99609375 *Command a value for the output, if Data Identification is set Data Link. In arbitrary engineering units.*
Analog Output #2	2	35	17	19	
PWM Output #1	2	37	19	21	Scaling: 1/256 / bit Offset: -8,224,768 Range: -8,224,768 to 8,224,767.99609375 *Command a value for the PWM output*
PWM Output #2	2	39	21	23	
PWM Output #3	2	41	23	25	

7.5.3 Read - Remote Inputs and Outputs

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Remote Digital Input #1 to Input #16	1	1825	1259	1621	0 = Deactivated 1 = Activated *Bit-0 to Bit-15: DI-1 to DI-16*
Remote Digital Input #17 to Input #32	1	1826	1260	1622	
Remote Digital Input #33 to Input #48	1	1827	1261	1623	
Remote Digital Output #1 to Output #16	1	1828	1262	1624	
Remote Digital Output #17 to Output #32	1	1829	1263	1625	
Remote Digital Output #33 to Output #48	1	1830	1264	1626	
Remote Analog Input #1 Sensor Value	1	1832	1265	1627	Scaling: 1/16 / bit Offset: -16 Range: -16 to 3999.9375 *Raw sensor value read from the input (mA)*
Remote Analog Input #2 Sensor Value	1	1833	1266	1628	
Remote Analog Input #3 Sensor Value	1	1834	1267	1629	
Remote Analog Input #4 Sensor Value	1	1835	1268	1630	
Remote Analog Input #5 Sensor Value	1	1836	1269	1631	
Remote Analog Input #6 Sensor Value	1	1837	1270	1632	
Remote Analog Input #7 Sensor Value	1	1838	1271	1633	
Remote Analog Input #8 Sensor Value	1	1839	1272	1634	

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Read - Remote Inputs and Outputs - continued

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Remote Analog Input #1 Data Value	2	1841	1273	1635	Scaling: 1/256 / bit Offset: -8,224,768 Range: -8,224,768 to 8,224,767.99609375 *Data value as read from the input, in engineering units.*
Remote Analog Input #2 Data Value	2	1843	1275	1637	
Remote Analog Input #3 Data Value	2	1845	1277	1639	
Remote Analog Input #4 Data Value	2	1847	1279	1641	
Remote Analog Input #5 Data Value	2	1849	1281	1643	
Remote Analog Input #6 Data Value	2	1851	1283	1645	
Remote Analog Input #7 Data Value	2	1853	1285	1647	
Remote Analog Input #8 Data Value	2	1855	1287	1649	
Remote Analog Output #1 Data Value	2	1857	1289	1651	
Remote Analog Output #2 Data Value	2	1859	1291	1653	
Remote Analog Output #3 Data Value	2	1861	1293	1655	
Remote Analog Output #4 Data Value	2	1863	1295	1657	
Remote Analog Output #5 Data Value	2	1865	1297	1659	
Remote Analog Output #6 Data Value	2	1867	1299	1661	
Remote Analog Output #7 Data Value	2	1869	1301	1663	
Remote Analog Output #8 Data Value	2	1871	1303	1665	
Remote Analog Output #1 Output Value	1	1872	1305	1667	Scaling: 1/16 / bit Offset: -16 Range: -16 to 3999.9375 *Raw sensor value read from the input (mA)*
Remote Analog Output #2 Output Value	1	1873	1306	1668	
Remote Analog Output #3 Output Value	1	1874	1307	1669	
Remote Analog Output #4 Output Value	1	1875	1308	1670	
Remote Analog Output #5 Output Value	1	1876	1309	1671	
Remote Analog Output #6 Output Value	1	1877	1310	1672	
Remote Analog Output #7 Output Value	1	1878	1311	1673	
Remote Analog Output #8 Output Value	1	1879	1312	1674	
Remote RTD #1 Sensor Value	1	1880	1313	1675	Scaling: 0.1 C/bit Offset: 200 Range: -200 to 850 C
Remote RTD #2 Sensor Value	1	1881	1314	1676	
Remote RTD #3 Sensor Value	1	1882	1315	1677	
Remote RTD #4 Sensor Value	1	1887	1316	1678	

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7.5.4 Write - Remote Outputs

Description TCP Port: 50200	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Remote Digital Output #1 to Output #48	1	43 to 90	37 to 84	39 to 86	0 = Deactivate 1 = Activate *Command to activate this output. Applies only when Usage Type is set to Data Link.*
Remote Analog Output #1	2	91	85	87	Scaling: 1/256 / bit Offset: -8,224,768 Range: -8,224,768 to 8,224,767.99609375 *Command a value for the output, if Data Identification is set to Data Link. In arbitrary engineering units.*
Remote Analog Output #2	2	93	87	89	
Remote Analog Output #3	2	95	89	91	
Remote Analog Output #4	2	97	91	93	
Remote Analog Output #5	2	99	93	95	
Remote Analog Output #6	2	101	95	97	
Remote Analog Output #7	2	103	97	99	
Remote Analog Output #8	2	105	99	101	

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8.0 Cat ECS Events

The Cat ECS controls provide both internal and J1939 data link transmitted events. The specifics of these various types of events with respect to SCADA Modbus are listed below with details in respective subsection:

- Events Acknowledge and Reset
- Events Acknowledgement Status
- Hard Shutdown, Soft Shutdown, Lamp, Horn and Display Events and Status
- User Alarm Group
- Annunciator-related Events
- Event Logs

8.1 Acknowledge and Reset

Description TCP Port: 50200	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Alarm Acknowledge Command	1	1	1	-	0 = FALSE 1 = TRUE
Reset All Events Command	1	2	2	-	

8.2 Acknowledge Status

Description TCP Port: 50201	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Alarm Active/Not Acknowledged	1	640	640	-	Indicates Alarm is active and not Acknowledged
Alarm Active/Acknowledged	1	641	641	-	Indicates Alarm active is Acknowledged
Shutdown Active	1	642	642	-	Indicates Shutdown is active
Shutdown Active/acknowledged	1	643	643	-	Indicates Shutdown active is acknowledged

8.3 Hard Shutdown, Soft Shutdown, Lamp, Horn and Display Events

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
System Event Lamp Status	1	2	2	-	Bits 15:4 Unused Bits 3:2 Amber Lamp Status (Boolean) Bits 1:0 Red Lamp Status (Boolean)
System Horn Active Status	1	3	3	-	0 = False 1 = True 2 = Error 3 = Disabled Or Not Available
Hard Shutdown Status	1	4	4	-	
Soft Shutdown Status	1	5	5	-	
On package Display Event Info	1	30	30	-	Bit 1: Horn Status Bits 2-3: Red Lamp Bits 4-5: Amber Lamp
Off package Display Event Info	1	31	31	-	

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8.4 User Alarm Groups (SCADA Alarm Bit Packing)

The Cat ECS contols provide a feature called User Alarm Group. Using this feature user can configure the desired alarm using SPN-FMI at the word and bit level. There are ten (10) user configured alarm group words provided per Cat ECS controller. This gives the user a total of 160 bits/alarms (16-bits per word x 10 words) to configure.

Leveraging this feature user can minimize the number of read requests by reading the required events in single transaction at the word level. Furthermore prioritizing and adjusting the polling rate of critical alarm words more frequently versus less-critical alarm words less frequently ensures data communication efficiency, and balance between real-time data needs and network load.

See section “SCADA Alarm Bit Packing” of Cat ECS 100 and Cat ECS 200 software operating manual on how to configure SCADA Alarm Bits.

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Alarm Group A	1	101	101	101	1 = Alarm ON (TRUE) 0 = Alarm OFF (FALSE) *Bit-0 to Bit-15: Event #1 to Event #16*
Alarm Group B	1	102	102	102	
Alarm Group C	1	103	103	103	
Alarm Group D	1	104	104	104	
Alarm Group E	1	105	105	105	
Alarm Group F	1	106	106	106	
Alarm Group G	1	107	107	107	
Alarm Group H	1	108	108	108	
Alarm Group I	1	109	109	109	
Alarm Group J	1	110	110	110	

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8.5 Annunciator

Description TCP Port: 50201	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	0 = FALSE, 1 = TRUE
Engine Auxiliary Switch Status (E-Stop) (Red)	1	611	611	-	Indicates SPN 970 is active
Generator not In Auto Warning (Amber)	1	612	612	-	Indicates SPN 3567 is active
Overcrank (Engine Failure to Start) (Red)	1	613	613	-	Indicates SPN 1664 is active
Overspeed (Red)	1	614	614	-	Indicates overspeed is active
EPS Supplying Load (Green)	1	615	615	-	Indicates SPN 3829 is active
Overload Alarm/Load Shed Contact (Amber)	1	616	616	-	Indicates overload alarm is active
High Coolant Temp (Amber)	1	617	617	-	Indicates SPN 110 is active
High Coolant Temp (Red)	1	618	618	-	Indicates SPN 110 is active
Low Coolant Temp Warning (Amber)	1	619	619	-	Indicates SPN 111 is active
Low Cranking Voltage Warning (Amber)	1	620	620	-	Indicates SPN 4256 is active
Air Damper Shutdown (Red)	1	621	621	-	Indicates SPN 3667 is active
High Battery Voltage (Amber)	1	622	622	-	Indicates SPN 168 is active
High Battery Voltage (Red)	1	623	623	-	Indicates SPN 168 is active
Low Battery Voltage Warning (Amber)	1	624	624	-	Indicates SPN 168 is active
Low Oil Press (Amber)	1	625	625	-	Indicates SPN 100 is active
Low Oil Press (Red)	1	626	626	-	Indicates SPN 100 is active
Low Coolant Level (Amber)	1	627	627	-	Indicates SPN 111 is active
Low Coolant Level (Red)	1	628	628	-	Indicates SPN 111 is active
Low Fuel Level (Amber)	1	629	629	-	Indicates SPN 96 is active
Low Fuel Level (Red)	1	630	630	-	Indicates SPN 96 is active
Battery Charger AC Failure (Amber)	1	631	631	-	Indicates SPN 4990 is active
SCR SPN 4792 (Amber)	1	632	632	-	Indicates SPN 4792 is active
SCR SPN 4792 (Red)	1	633	633	-	Indicates SPN 4792 is active
Custom Event 1 (Amber)	1	634	634	-	Indicates custom event warning is active
Custom Event 1 (Red)	1	635	635	-	Indicates custom event 1 shutdown is active
Custom Event 1 (Green)	1	636	636	-	Indicates custom event 1 Condition exists is active
Custom Event 2 (Amber)	1	637	637	-	Indicates custom event 2 shutdown is active
Custom Event 2 (Red)	1	638	638	-	Indicates custom event 2 Condition exists is active
Custom Event 2 (Green)	1	639	639	-	Indicates custom 2 event shutdown is active

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8.6 Event Logs

Cat ECS 100 and 200 provides information on both internal events and those transmitted over the J1939 data link by various installed modules. Details of both Cat ECS Controller and supported additional module events can be read over SCADA Modbus.

Cat ECS supports up to 40 log entries at a time for all the system event logs and up to 500 logs for each status event as mentioned above. Each Log Entry consists of 14 Modbus Registers. Once the event log or status event log is filled, old events are lost and are replaced by the most recent events.

The events can be viewed via Cat ET (Cat Service Tool) by selecting the Logged Diagnostic Codes button on the toolbar or the HMI display under event logs menu. It is recommended to consult HMI and Cat ECS software operating manuals for further information on Event Logs and Diagnostics.

Use Cat ET or HMI to view event logs. However, if event log monitoring over SCADA Modbus is crucial and highly desirable, then follow data prioritization and selective polling methodolgy presented in the “Best Practices” section of this manual to ensure communication, and overall system performance and stability.

8.6.1 Event Logs – Number of Active Events

Description TCP Port: 50201	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Number of Active Events - Genset Controller	1	16	16	-	Scaling: 1 / bit Offset: 0 Range: 0 to 65535 Scaling: 1 / bit Offset: 0 Range: 0 to 64255
Number of Active Events - Paralleling Controller	1	-	-	2	
Number of Active Events - Engine Controller	1	17	17	-	
Number of Active Events - Secondary Engine Controller	1	18	18	-	
Number of Active Events - External I/O 1 Controller	1	19	19	-	
Number of Active Events - External I/O 2 Controller	1	20	20	-	
Number of Active Events - External I/O 3 Controller	1	21	21	-	
Number of Active Events - External I/O 4 Controller	1	22	22	-	
Number of Active Events - Digital AVR Controller	1	23	23	-	
Number of Active Events - RTD Controller	1	24	24	-	
Number of Active Events - Thermocouple 1 Controller	1	25	25	-	
Number of Active Events - Thermocouple 2 Controller	1	26	26	-	
Number of Active Events - SCR Controller	1	27	27	-	
Number of Active Events - ISM Controller	1	28	28	-	

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Event Logs – Number of Active Events - continued

Description TCP Port: 50201	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Number of Active Events - ICPS Controller	1	29	29	-	Scaling: 1 / bit Offset: 0 Range: 0 to 65535 Scaling: 1 / bit Offset: 0 Range: 0 to 64255
Number of Status Events in Log – Genset Controller	1	644	644	-	
Number of Status Events in Log – Paralleling Controller	1	-	-	563	
Number of LSLD Status Events in Log	1	-	-	2568	

8.6.2 Event Logs – System Events (Genset and Engine)

Description TCP Port: 50201	Word Count	Modbus Register		Register Information
		100G	200G	
Genset Event Log Entry (#1 … #40)				Register 13 = Log Entry Index (J1939 Count) Register 12:11 = SPN/FMI (Event ID) Register 10 = Flags / Count (Event Flags) Register 9:8 = First Hourmeter (Event Log Hourmeter) Register 7:6 = Last Hourmeter (Event Log Hourmeter) Register 5:3 = First Timestamp (Time/Date) Register 2:0 = Last Timestamp (Time/Date)
Genset Event Log Entry 1	14	51	51	
Genset Event Log Entry 40	14	597	597	
Engine Event Log Entry (#1 … #40)				
Engine Event Log Entry 1	14	2650	2650	
Engine Event Log Entry 40	14	4879	4879	

Note: Refer to Cat ECS 100 v1.4 and Cat ECS 200 v1.1 SCADA evetns register lists spreadsheets for complete list of all available modules event log registers.

8.6.3 Event Logs – Status Events

Description TCP Port: 50201	Word Count	Modbus Register		Register Information
		100G	200G	
Current Date/Time				
Status Event 0 Event ID (Snapshot Time)	1	645	645	Log Entry Index (J1939 Count) – Current Time
Status Event 0 Snapshot Timestamp	3	646	646	Register 2:0 = Timestamp (Time/Date)
Status Event 499 Event ID	1	2645	2645	Log Entry Index (J1939 Count)
Status Event 499 Timestamp	3	2646	2646	Register 2:0 = Timestamp (Time/Date)

Note: Refer to Cat ECS 100 v1.4 and Cat ECS 200 v1.1 SCADA evetns register lists spreadsheets for complete list of all available modules event log registers.

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9.0 Network and Module

Note: The Modbus data tables in this manual are based on the Cat ECS 100 v1.4 and Cat ECS 200 v1.1 register lists spreadsheets. These versions of the Cat ECS 100 and Cat ECS 200 Modbus register lists are current as of the time of the publication of this manual.

9.1 Network Status

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Primary Data Link Status	1	1050	700	350	Indicates any present faults on CAN datalink. *00 = Ok* *01 = Fault* *11 = Disabled Or Not Available*
Accessory Data Link Status	1	1051	701	-	
Ethernet 2-Wire Port A Status	1	1280	930	577	Indicates Enable status of Cat ECS Ethernet ports *0 = Disabled* *1 = Enabled*
Ethernet 2-Wire Port B Status	1	1281	931	578	
Ethernet 4-Wire Port #1 Status	1	1282	932	579	
Ethernet 4-Wire Port #2 Status	1	1283	933	580	
Ethernet 4-Wire Port #3 Status	1	1284	934	581	
Ethernet 4-Wire Port #4 Status	1	1285	935	582	

9.2 Module Online Status

Description TCP Port: 50201	Word Count	Modbus Register			Scaling, Offset, Range (Status / State / Info)
		100G	200G	200P	0 = FALSE (Offline), 1 = TRUE (Online)
Genset Control Online	1	1	1	-	Indicates whether the Genset controller GSC (Cat ECS) is on the communication network. This will always be True.
Genset Paralleling Control Online	1	-	-	1	Indicates whether the Genset Paralleling controller GSC (Cat ECS) is on the communication network. This will always be True.
Engine Control Online	1	2	2	-	Indicates whether the engine ECM is in communication with the Genset controller GSC (Cat ECS) via CAN as detected by Genset controller
Secondary Engine Control Online	1	3	3	-	Indicates whether the secondary engine ECM is in communication with the Genset controller GSC (Cat ECS) via CAN as detected by Genset controller
External I/O #1 Online	1	4	4	-	Indicates whether the DIO module instance #1 is in communication with the Genset controller GSC (Cat ECS) via CAN as detected by Genset controller
External I/O #2 Online	1	5	5	-	Indicates whether the DIO module instance #2 is in communication with the Genset controller GSC (Cat ECS) via CAN as detected by Genset controller
Digital AVR Online	1	6	6	-	Indicates whether the digital AVR module is in communication with the Genset controller GSC (Cat ECS) via CAN as detected by Genset controller
RTD Module Online	1	7	7	-	Indicates whether the RTD module is in communication with the Genset controller GSC (Cat ECS) via CAN as detected by Genset controller
Thermocouple #1 Online	1	8	8	-	Indicates whether the thermocouple module instance #1 is in communication with the Genset controller GSC (Cat ECS) via CAN as detected by Genset controller

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Module Online Status - continued

Description TCP Port: 50201	Word Count	Modbus Register			Scaling, Offset, Range (Status / State / Info)
		100G	200G	200P	0 = FALSE (Offline), 1 = TRUE (Online)
Thermocouple #2 Online	1	9	9	-	Indicates whether the thermocouple module instance #2 is in communication with the Genset controller GSC (Cat ECS) via CAN as detected by Genset controller
SCR Module Online	1	10	10	-	Indicates whether the SCR module is in communication with the Genset controller GSC (Cat ECS) via CAN as detected by Genset controller
Telematics Module Online	1	11	11	-	Indicates whether the PLE telematics module is online
External I/O #3 Online	1	12	12	-	Indicates whether the DIO module instance #3 is in communication with the Genset controller GSC (Cat ECS) via CAN as detected by Genset controller
External I/O #4 Online	1	13	13	-	Indicates whether the DIO module instance #4 is in communication with the Genset controller GSC (Cat ECS) via CAN as detected by Genset controller
ISM Module Online	1	14	14	-	Indicates whether the ISM module is online
ICPS Module Online	1	15	15	-	Indicates if the In Cylinder Pressure Sensor (ICPS) module is online as detected by the Genset controller

9.3 Module Information

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Power System Control Unit Type (Standby) A6L3 Device Info	1	1350	950	600	Indicates Cat ECS control unit type *Device ID MID:* *$164 - Single Genset Control* *$168 - Genset Control* *$169 - Genset Paralleling Control*
Power System Software Group Number A6L3 Device Info	5	1351	951	601	Indicates Power System Software Group *0x00 to 0x7F for each byte (char)*
Power System Software Version A6L3 Device Info	13	1356	956	606	Indicates Power System Software Version *0x00 to 0x7F for each byte (char)*
Power System Control Hardware Part Number A6L3 Device Info	5	1369	969	619	Indicates Power System Control Hardware Part Number *0x00 to 0x7F for each byte (char)*
Power System Control Serial Number A6L3 Device Info	8	1374	974	624	Indicates Power System Control Serial Number *0x00 to 0x7F for each byte (char)*
Power System Control Unit Number A6L3 Device Info	1	1382	982	632	Indicates Power System Control Unit Number A6L3 Device Info *Scaling: 1/bit, Offset: 0, Range: 0 to 65535*
Engine ID/Model	9	1383	983	-	Indicates Engine Equipment ID *0x00 to 0x7F for each byte (char)*
Engine Serial Number	4	1392	992	-	Used to read the engine serial number *0x00 to 0x7F for each byte (char)*
Software Group Part Number Engine Device Info	25	1396	996	-	Engine Software Group Part Number *0x00 to 0x7F for each byte (char)*
Personality Module Part Number Engine Device Info	5	1421	1021	-	Engine Personality Module Part Number *0x00 to 0x7F for each byte (char)*

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Module Information - continued

Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Engine Rating Map #	1	1426	1026	-	Engine Rating Map Number *Scaling: 1/bit, Offset: 0, Range: 0 to 65535*
Telematics Cellular Packet Data Protocol (PDP) Context Activation Status	1	1427	1217	-	Indicates the cellular packet data protocol context activation status of the cellular device data network as detected by the PLE telematics device when connected *$000E = Active* *$000F = Inactive* *$04C8 = Missing or Unknown APN*
Number of GPS Satellites	1	1428	1218	-	Indicates the number of GPS Satellites as detected by the PLE telematics device *Scaling: 1/bit, Offset: 0, Range: 0 to 65535*
Telematics Cellular Signal Quality	1	1429	1219	-	Indicates the strength of the signal being received by the cellular device as detected by the PLE telematics device when connected *Scaling:1% / bit, Offset:0, Range:0 to 250%*
Telematics Satellite Communication Status	1	1430	1220	-	Indicates the status of a communications channel with the currently selected satellite as detected by the PLE telematics device when connected *$022C = Established* *$022D = Not Established*
Telematics Communications Modem (Satellite) Signal Quality	1	1431	1221	-	Indicates the quality of the signal being received by the communications modem as detected by the PLE telematics device when connected *Scaling:1% / bit, Offset:0, Range: 0 to 250%*

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10.0 Grid Code

Note: The Modbus data tables in this manual are based on the Cat ECS 100 v1.4 and Cat ECS 200 v1.1 lists spreadsheets. These versions of the Cat ECS 100 and Cat ECS 200 Modbus register lists are current as of the time of the publication of this manual.

Description TCP Port: 502	Word Count	Modbus Register 200P	Scaling, Offset, Range / Status / State / Info
Generator Desired kVAr	2	645	Scaling: 1 VAr / bit Offset: -2000000000 VAr Range: -2000000000 to +2211081215 VAr
Voltage Compensation Bias Percent	1	653	Scaling: 0.0078125 % / bit Offset: -251 Range: -251 to 250.99 %
Real Power Active Setpoint Source	1	1733	0000 = Genset Operator 0001 = Service Tool 0010 = Third Party Vendor 0011 = Utility Operator 0100 = Limited Frequency Sensitive Mode Overfrequency 0101 = Limited Frequency Sensitive Mode Underfrequency 0110 = Utility Operator Limitation 0111 = Utility Parallel Voltage-Real Power Curve 1000 - 1101 = Reserved 1110 = Error 1111 = Not Available
Primary Real Power Active Setpoint Offset Source	1	1734	00 = FSM 01 = None 10 = Error 11 = Not Available
Secondary Real Power Active Setpoint Offset Source	1	1735	00 = ILF Regulation 01 = Secondary Regulation 10 = Error 11 = None
Real Power Setpoint Percentage	1	1736	Scaling: 0.01 % / bit Offset: -321.27 % Range: -321.27 to 321.28 %
Generator Real Load Control Base Load Setpoint	1	1737	
Reason of limitation	1	1738	Utility Operator Limitation Level 1 Utility Operator Limitation Level 2 Utility Operator Limitation Level 3 Utility Operator Limitation Level 4 Utility Operator Limitation P(U)
Real Power limitation	1	1739	Scaling: 0.4 % / bit Offset: 0 % Range: 0.0 to 100.0 %

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Grid Code - continued

Description TCP Port: 502	Word Count	Modbus Register 200P	Scaling, Offset, Range / Status / State / Info
Real Power Ramp Rate Source	1	1740	Genset Operator Third Party Operator Utility Operator Demand LFSM Stage #1 LFSM Stage #2 Disconnection from Utility P(U) FSM Utility Operator Limitation Genset Normal Starting Reconnection after Utility Decoupling Error Not Available
Primary Real Power Offset Ramp Rate Source	1	1741	FSM Increase FSM Decrease None Error Not Available
Secondary Real Power Offset Ramp Rate Source	1	1742	ILF Regulation Secondary Regulation Error None
Real Power Increase Ramp Rate Setpoint	1	1743	Scaling: 0.01 % / s / bit Offset: 0 % /s Range: 0 to 642.55 % / s
Real Power Decrease Ramp Rate Setpoint	1	1744	
Real Power Ramp Rate Limit Status	1	1745	Not Actively Limiting Actively Limiting Error Not Available
Real Power Ramp Rate Limit Enable Status	1	1746	Disabled Enabled Error Not Available
Third Party Operator Real Power Status	1	1747	Not Actively Controlling Actively Controlling Error Not Available
Third Party Operator Real Power Enable Status	1	1748	Disabled Enabled Error Not Available
Third Party Operator Real Power Setpoint	1	1749	Scaling: 0.01 % / bit Offset: -321.27 % Range: -321.27 to 321.28 %
Utility Operator Real Power Status	1	1750	Not Actively Limiting Actively Limiting Error Not Available

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Grid Code - continued

Description TCP Port: 502	Word Count	Modbus Register 200P	Scaling, Offset, Range / Status / State / Info
Utility Operator Real Power Enable Status	1	1751	Disabled Enabled Error Not Available
Utility Operator Real Power Percentage Setpoint	1	1752	Scaling: 0.01 % / bit Offset: -321.27 % Range: -321.27 to 321.28 %
Utility Operator Real Power Export Limit Status	1	1753	Inactive Active Error Not Available
Utility Operator Real Power Export Limit Percentage	1	1754	Scaling: 0.01 % / bit Offset: -321.27 % Range: -321.27 to 321.28 %
Utility Operator Real Power Export Limit Setpoint 1 Status	1	1755	Not Actively Limiting Actively Limiting Error Not Available
Utility Operator Real Power Export Limit Setpoint 2 Status	1	1756	
Utility Operator Real Power Export Limit Setpoint 3 Status	1	1757	
Utility Operator Real Power Export Limit Setpoint 4 Status	1	1758	
Limited Frequency Sensitive Mode Status	1	1759	Inactive Active Error Not Available
Limited Frequency Sensitive Mode Over frequency Enable Status	1	1760	Disabled Enabled Error Not Available
Limited Frequency Sensitive Mode Under frequency Status	1	1761	
Limited Frequency Sensitive Mode Over frequency Enable Status Deactivation Threshold Reset Status	1	1762	Reset Not Requested Reset Requested Error Not Available
Limited Frequency Sensitive Mode Real Power Setpoint	1	1763	Scaling: 0.01 % / bit Offset: -321.27 % Range: -321.27 to 321.28 %
Limited Frequency Sensitive Mode Event	1	1764	None LFSM O LFSM U LFSM Stage 2
Limited Frequency Sensitive Mode Starting Power	1	1765	Scaling: 1 / 128 Hz/bit Offset: 0 Range: 0.0 to 100.0
Utility Parallel Voltage-Real Power Curve Status	1	1766	Inactive Active Error Not Available

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Grid Code - continued

Description TCP Port: 502	Word Count	Modbus Register 200P	Scaling, Offset, Range / Status / State / Info
Utility Parallel Voltage-Real Power Curve Setpoint Percentage	1	1767	Scaling: 0.01 % / bit Offset: -321.27 % Range: -321.27 to 321.28 %
Frequency Sensitive Mode Status	1	1768	Inactive Active Error Not Available
Frequency Sensitive Mode Enable Status	1	1769	Disabled Enabled Error Not Available
Frequency Sensitive Mode Reservation Active	1	1770	Inactive Active Error Not Available
Frequency Sensitive Mode Real Power Setpoint	1	1771	Scaling: 0.01 % / bit Offset: -321.27 % Range: -321.27 to 321.28 %
Frequency Sensitive Mode Active Power Applied	1	1772	
Secondary Frequency Control Status	1	1773	Inactive Active Error Not Available
Integral Local Frequency Control and Secondary Frequency Control Status	1	1774	Disabled Enabled Error Not Available
Secondary Frequency Control Setpoint Reserve Active	1	1775	Inactive Active Error Not Available
Desired Secondary Frequency Control Real Power Offset	1	1776	Scaling: 0.01 % / bit Offset: -321.27 % Range: -321.27 to 321.28 %
Secondary Frequency Control Setpoint Power Applied	1	1777	
Integral Local Frequency Control Status	1	1778	Inactive Active Error Not Available
Integral Local Frequency Control Reset Status		1780	Reset not requested Reset requested Error Not Available
Integral Local Frequency Control Real Power Offset	1	1781	Scaling: 0.01 % / bit Offset: -321.27 % Range: -321.27 to 321.28 %
Integral Local Frequency Control Real Power Applied	1	1782	Scaling: 0.1 % / bit Offset: 0 % Range: 0 to 100 %

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Grid Code - continued

Description TCP Port: 502	Word Count	Modbus Register 200P	Scaling, Offset, Range / Status / State / Info
Real Power Reversion Status	1	1783	Inactive Active Error Not Available
Real Power and Reactive Power Reversion Status	1	1784	
Reactive Power Active Setpoint Source	1	1793	None Power Factor Var Var(Volt)-Var_Shift Var(Volt)-Volt_Shift Var(kW) Power Factor(kW) Power Factor(Volt) Volt Reserved Error Not available
Reactive Power Active Demanded Setpoint Source	1	1794	
Reactive Power at Low Real Power Status	1	1795	Inactive Active Error Not Available
Reactive Power Reversion Status	1	1796	
Generator Under Excitation Limit Status	1	1797	
Generator Over Excitation Limit Status	1	1798	
Reactive Power Setpoint	1	1799	Scaling: 0.01 % / bit Offset: -321.27 % Range: -321.27 to 321.28 %
Reactive Power Value	1	1800	Scaling: 0.1 kVAr / bit Offset: -3212.7 kVAr Range: -3212.7 to 3212.8 %
Demanded Power Factor	1	1801	Scaling: 1/16384 / bit Offset: -1 Range: -1.0 to 1.0
Demanded Voltage	1	1802	Scaling: 0.01 % / bit Offset: -321.27 % Range: -321.27 to 321.28 %
Target Percent kVAr	1	1803	Scaling: 0.0078125 % / bit Offset: -250 % Range: -250 to 250.99 %
Target Percent kW	1	1804	Scaling: 0.0078125 % / bit Offset: -250 % Range: -250 to 250.99 %
Test Functions for the Protections Undervoltage Overvoltage Status	1	1987	

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Grid Code - continued

Description TCP Port: 502	Word Count	Modbus Register 200P	Scaling, Offset, Range / Status / State / Info
Generator Real Load Control State	1	1805	0 = Load Share - Disabled 1 = Load Share - Inactive 2 = Load Share - Soft Load 3 = Load Control - Soft Load 4 = Load Share - Soft Unload 5 = Load Control - Soft Unload 6 = Load Share - Failsafe Droop 7 = Load Share - Failsafe Isoch 8 = Load Control - Base Load 9 = Load Control - PF Control 11 = Load Share - MGDL/APCDL
Fault Ride Through Status	1	1980	0 = No_Event 1 = Fault_Ride_Through 2 = High_Voltage_Ride_Through 3 = Low_Voltage_Ride_Through
Test Function for the Simulated Utility Frequency Status	1	1982	Inactive Active Error Not Available
Simulated Utility Frequency Offset	1	1983	Scaling: 0.01 Hz / bit Offset: -321.27 Hz Range: -321.27 to 321.28 Hz
Simulated Utility Frequency Result	1	1984	Scaling: 1/128 Hz / bit Offset: 0 Range: 0 to 501.9922 Hz
Simulated Utility Frequency Deactivation Timer	1	1985	Scaling: 1 hour / bit Offset: 0 Range: 0 to 65503 hr
Test Functions for the Protections Utility Decoupling Status	1	1986	Inactive Active Error Not Available
Test Functions for the Protections Undervoltage Overvoltage Status	1	1987	

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11.0 Balance of Plant

Note: The Modbus data tables in this manual are based on the Cat ECS 100 v1.4 and Cat ECS 200 v1.1 lists spreadsheets. These versions of the Cat ECS 100 and Cat ECS 200 Modbus register lists are current as of the time of the publication of this manual.

Description TCP Port: 502	Word Count	Modbus Register 200P	Scaling, Offset, Range / Status / State / Info
Heating Circuit Supply Outlet Coolant Temperature	1	1692	Scaling: 0.03125 C / bit Offset: -273 Range: -273 to 1735 C
Heating Circuit Return Coolant Temperature	1	1693	
Engine Coolant Temperature 2	1	1694	
Engine Radiator 1 Control Temperature	1	1695	
Engine Aftercooler Coolant Temperature	1	1696	
Engine Radiator 2 Control Temperature	1	1697	
Auxiliary Temperature 1	1	1698	Scaling: 1 C / bit Offset: -40 Range: -40 to 210 C
Auxiliary Temperature 2	1	1699	
Auxiliary Temperature 3	1	1700	
Genset Enclosure Intake Air Temperature	1	1701	
Genset Enclosure Outlet Air Temperature	1	1702	
Heating Circuit Pump Control	1	1703	00 = False 01 = True 10 = Error 11 = Disabled Or Not Available
Engine Coolant Pump Control	1	1704	
Engine Coolant Pump 2 Control	1	1705	
Engine Exhaust Heat Exchanger Outlet Temperature	1	1706	Scaling: 0.03125 C / bit Offset: -273 Range: -273 to 1735 C
Engine Exhaust Heat Exchanger Bypass Actuator 1 Desired Position	1	1707	Scaling: 0.4 % / bit Offset: 0 Range: 0 to 100%
Engine Exhaust Heat Exchanger Bypass Actuator 1 Position	1	1708	
Desired Engine Coolant Temperature	1	1709	Scaling: 0.03125 C / bit Offset: -273 Range: -273 to 1735 C
Desired Heating Circuit Thermostat Opening	1	1710	Scaling: 0.4 % / bit Offset: 0 Range: 2 to 100%
Heating Circuit Thermostat Opening	1	1711	Scaling: 0.4 % / bit Offset: 0 Range: 3 to 100%
Desired Engine Coolant Thermostat Opening	1	1712	Scaling: 0.4 % / bit Offset: 0 Range: 4 to 100%
Engine Coolant Thermostat Opening	1	1713	Scaling: 0.4 % / bit Offset: 0 Range: 5 to 100%
Desired Engine Radiator 1 Control Temperature	1	1714	Scaling: 0.03125 C / bit Offset: -273 Range: -273 to 1735 C

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Balance of Plant - continued

Description TCP Port: 502	Word Count	Modbus Register 200P	Scaling, Offset, Range / Status / State / Info
Engine Radiator Temperature Control Selection Status	1	1715	00 = False 01 = True 10 = Error 11 = Disabled Or Not Available
Engine Fan 1 Requested Percent Speed	1	1716	Scaling: 0.4 % / bit Offset: 0 Range: 6 to 100%
Engine Aftercooler Coolant Pump Control	1	1717	00 = False 01 = True 10 = Error 11 = Disabled Or Not Available
Engine Desired Aftercooler Coolant Intake Temperature (High Resolution)	1	1718	Scaling: 0.03125 C / bit Offset: -273 Range: -273 to 1735 C
Engine Desired Aftercooler Coolant Thermostat Opening	1	1719	Scaling: 0.4 % / bit Offset: 0 Range: 7 to 100%
Engine Charge Air Cooler Thermostat Opening	1	1720	Scaling: 0.4 % / bit Offset: 0 Range: 8 to 100%
Desired Engine Radiator 2 Control Temperature	1	1721	Scaling: 0.03125 C / bit Offset: -273 Range: -273 to 1735 C
Engine Fan 2 Requested Percent Speed	1	1722	Scaling: 0.4 % / bit Offset: 0 Range: 9 to 100%
Genset Enclosure Ventilation Louver Actuator 1 Position	1	1724	Scaling: 0.4 % / bit Offset: 0 Range: 0 to 100%
Genset Enclosure Ventilation Louver Actuator 2 Position	1	1725	
Genset Enclosure Ventilation Recirculation Louver Actuator Position	1	1726	Scaling: 0.4 % / bit Offset: 0 Range: 0 to 100%
Genset Enclosure Fan Requested Percent Speed	1	1728	
Engine Exhaust Temp from base app IO	1	1729	Scaling: 0.03125 C / bit Offset: -273 Range: -273 to 1735 C
After Treatment Gas Oxidation Catalyst outlet Temperature	1	1730	
Engine Fan 1 Requested Number of Fan Groups	1	1731	Scaling: 1 / bit Offset: 0 Range: 0 to 65535
Engine Fan 2 Requested Number of Fan Groups	1	1732	

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12.0 Examples

12.1 SCADA Password Write

This example illustrates how to write SCADA password **12345678** to Cat ECS 100 and Cat ECS 200 Genset controllers. The following tables illustrate Cat ECS 100 and Cat ECS 200 SCADA password registers, format and conversion:

Steps:

- Convert SCADA Modbus password from ASCII to HEX at the **Byte** level. See ASCII table below. Password characters are highlighted for user convenience.
- Convert corresponding HEX values to Decimal at the **Word** level if the password is desired to be sent in decimal. For example**,** 0x3132 HEX word value converts to 12594 Decimal.
- Fill up remaining bytes of the password to 0x20 HEX (ASCII Space).
- Place the password values at the Modbus registers shown below.
- Send the Eight (8)-Word password write command to Cat ECS 100 or Cat ECS 200 Genset Controller in either HEX or Decimal.

Modbus Register - Cat ECS 100 Genset TCP Port: 50200	10	11	12	13	14	15	16	17
ASCII Password	12	34	56	78	SPSP	SPSP	SPSP	SPSP
HEX	3132	3334	3536	3738	2020	2020	2020	2020
DEC	12594	13108	13622	14136	8224	8224	8224	8224
Modbus Register - Cat ECS 200 Genset TCP Port: 50200	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
ASCII Password	12	34	56	78	SPSP	SPSP	SPSP	SPSP
HEX	3132	3334	3536	3738	2020	2020	2020	2020
DEC	12594	13108	13622	14136	8224	8224	8224	8224

Hex	Value	Hex	Value	Hex	Value	Hex	Value	Hex	Value	Hex	Value	Hex	Value	Hex	Value
00	NUL	10	DLE	20	SP	30	0	40	@	50	P	60	`	70	p
01	SOH	11	DC1	21	!	31	1	41	A	51	Q	61	a	71	q
02	STX	12	DC2	22	"	32	2	42	B	52	R	62	b	72	r
03	ETX	13	DC3	23	#	33	3	43	C	53	S	63	c	73	s
04	EOT	14	DC4	24	$	34	4	44	D	54	T	64	d	74	t
05	ENQ	15	NAK	25	%	35	5	45	E	55	U	65	e	75	u
06	ACK	16	SYN	26	&	36	6	46	F	56	V	66	f	76	v
07	BEL	17	ETB	27	'	37	7	47	G	57	W	67	g	77	w
08	BS	18	CAN	28	(	38	8	48	H	58	X	68	h	78	x
09	HT	19	EM	29	)	39	9	49	I	59	Y	69	i	79	y
0A	LF	1A	SUB	2A	*	3A	:	4A	J	5A	Z	6A	j	7A	z
0B	VT	1B	ESC	2B	+	3B	;	4B	K	5B	[	6B	k	7B	{
0C	FF	1C	FS	2C	,	3C	<	4C	L	5C	\	6C	l	7C	|
0D	CR	1D	GS	2D	-	3D	=	4D	M	5D	]	6D	m	7D	}
0E	SO	1E	RS	2E	.	3E	>	4E	N	5E	^	6E	n	7E	~
0F	SI	1F	US	2F	/	3F	?	4F	O	5F	_	6F	o	7F	DEL

Figure 12.1: ASCII Table

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12.2 Engine Speed Monitor and Control

The Cat ECS controller can take the desired speed bias from any of the following sources:

- SCADA (MODBUS) command
- Display buttons
- Digital Input
- Analog Input
- Preset value based on Rated/Idle speed setpoints

It is recommended to read Cat ECS 100 and Cat ECS 200 software operating manual section “Engine Speed” and “Idle/Rated Speed” to understand the overall aspect of Speed control via the above listed input sources.

This section lists the SCADA Modbus data points that need to be taken into consideration to monitor and control Engine Speed via SCADA Modbus.

- **Idle Switch Position and Status**
- **Engine Speed**
- **Total Speed Bias Percent** - The bias being applied to the speed command output, as a percentage of 200 rpm. This will change by 0.5% each time the Speed Adjust left/right arrow is pressed on the Genset Control display.
  Note: The total speed bias is a percentage of 200 rpm. This will increase or decrease the speed by 0.5% each time the Speed is adjusted from listed sources, or any time the Speed Bias Percent Increment is written. The min/max values for this data are ± 100%.
- **Desired Engine Speed** - Engine speed currently being requested by the Genset Control.
- **Manual Speed Bias Percent** - Manual speed bias from all manual speed adjustment sources (display, Modbus NOT including Analog In) as a percentage of nominal voltage.
- **Speed Bias Percent Increment Command** - This WRITE command Increments the percentage bias being applied to the speed command output. This value gets added to the Speed Bias Percent register (the bias decreases if a negative value is written).

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Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Engine Speed	1	165	159	-	Scaling: 0.125 rpm / bit Offset: 0 Range: 0 to 8031.875 rpm
Desired Engine Speed	1	410	-	240	
Desired Engine Speed	1	546	333	-	
Total Speed Bias Percent	1	1454	-	654	Scaling: 0.0078125 % / bit Offset: -251 Range: -251 to 250.99 %
Manual Speed Bias Percent	1	1455	-	655	
Desired Speed Bias via Cat ECS Local or Remote I/O	1	1456	-	656	
Engine Low Idle Switch Position *Indicates whether the Engine is in low idle state*	1	533	323	-	0 = False 1 = True 2 = Error 3 = Disabled Or Not Available
Force Idle Status (From any source) *Indicates if the Cat ECS is in idle mode regardless of source that triggers the forced idle.*	1	1320	1052	-	
Idle/Rated Status Switch Enable Status *Indicates status whether the force idle feature is enabled. The force idle feature commands the engine to run at idle, based on user input.*	1	1321	-	-	$000C = Enabled $000D = Disabled
Description TCP Port: 50200	**Word Count**	**Modbus Register**			**Scaling, Offset, Range / Status / State / Info**
		100G	**200G**	**200P**	
Speed Bias Percent Increment Command *Increments the percentage bias being applied to the speed command output. This increment gets applied to the total speed*	1	4	-	2	Scaling: 0.0078125 % / bit Offset: -251 Range: -251 to 250.99 %

Speed Bias Percent Increment:

One (1) Word Raw Value (WRITE) = (Engineering Value - Offset) / Scaling

Speed Bias Percent Increment Command = (Engineering Value – (-251)) / 0.0078125

- **Genset Rated Speed:** 1800 rpm
- **Genset Running RPM:** 1750 rpm

  Speed Bias Percent Increment (%): ((1800 – 1750) / 1800) x 100 = 2.78%
  Y (2.78) = (2.78 – (-251)) / 0.0078125 = **32484**
- **Genset Rated Speed:** 1800 rpm
- **Genset Running RPM:** 1825 rpm

  Speed Bias Percent Increment (%): ((1800 – 1825) / 1800) x 100 = -1.39%
  Y (-1.39) = (-1.39 – (-251)) / 0.0078125 = **31950**

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12.3 Genset Voltage Monitor and Control

The Cat ECS controller can take the desired voltage bias value from any of the following sources:

- SCADA (MODBUS) command
- Display buttons
- Digital Input
- Analog Input

If multiple inputs are active at the same time, the total bias is the sum of all inputs, regardless of if any inputs are negative and others are positive.

It is recommended to read Cat ECS 100 and Cat ECS 200 software operating manual section "Generator Output Voltage" to understand the overall aspect of Generator Voltage Output control via the above listed input sources.

This section lists the SCADA Modbus data points needed to take into consideration to monitor and control the Genset Voltage via SCADA Modbus.

- **Generator Average AC RMS Voltage**
- **Total Voltage Bias Percent** - The bias being applied to the voltage regulator desired output voltage, as a percentage of rated voltage (-100% to +100%). This value will increase or decrease by 0.5% each time the Voltage is adjusted Adjust via Cat ECS HMI, or any time the Voltage Bias percent increment/decrement Command is sent.
- **Desired Genset Output Voltage** - The desired Generator RMS voltage to be delivered by the genset.
- **Manual Voltage Bias Percent** - Manual voltage bias from all manual voltage adjustment sources (display, Modbus NOT including Analog In) as a percentage of nominal voltage.
- **Voltage Compensation Bias Percent** - Voltage bias as a percentage of nominal applied based on Reactive Droop or Line Loss Compensation settings.
- **Desired Generator Voltage Bias via Cat ECS Local or Remote I/O** - Desired Generator Voltage Bias via Cat ECS Local or Remote I/O.
- **Integrated Voltage Regulator (IVR) Operating State** - Indicates the operating state of the Integrated Voltage Regulator feature when configured.
- **Generator Excitation Command Percentage** - Generator excitation command percentage output from the Integrated Voltage Regulator feature (proportional to generator excitation current.
- **Generator Excitation Field Voltage from Data Link**
- **Generator Excitation Field Current from Data Link**
- **Voltage Bias Percent Increment WRITE Command** - Increments the percentage bias to be applied to the Automatic Voltage Regulator. This value gets added to the Total Voltage Bias Percent register (the bias decreases if a negative value is written).

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Description TCP Port: 502	Word Count	Modbus Register			Scaling, Offset, Range / Status / State / Info
		100G	200G	200P	
Voltage Compensation Bias Percent	1	1453	-	653	Scaling: 0.0078125 % / bit Offset: -251 Range: -251 to 250.99 %
Total Voltage Bias Percent	1	1452	-	652	
Manual Voltage Bias Percent	1	1455	-	650	
Desired Generator Voltage Bias via Cat ECS Local or Remote I/O	1	1451	-	651	
Generator Excitation Command Percentage	1	296	-	194	
Desired Genset Output Voltage	1	1447	-	648	Scaling: 1 V / bit Offset: 0 Range: 0 to 64255 V
Integrated Voltage Regulator (IVR) Operating State	1	1448	-	649	0 = Undefined 1 = Volts/Hz 2 = V/Hz + Droop 3 = V/Hz + Line Loss 4 = Soft Start 5 = Transient Relief Active 6 = Soft Recovery 7 = IVR Locked Out 8 = Synchronizing 9 = PF Control 10 = kVAr Sharing 11 = Stopping
Generator Excitation Field Voltage from Data Link	1	294	-	192	Scaling: 0.05 V / bit Offset: -1606 Range: -1606.0 to 1606.75 V
Generator Excitation Field Current from Data Link	1	295	-	193	Scaling: 0.05 A / bit Offset: 0 Range: 0 to 3212.75 A
Description TCP Port: 50200	**Word Count**	**Modbus Register**			**Scaling, Offset, Range / Status / State / Info**
		100G	**200G**	**200P**	
Voltage Bias Percent Increment Command *Increments the percentage bias being applied to the voltage command output. This increment gets applied to the total speed*	1	3	-	1	Scaling: 0.0078125 % / bit Offset: -251 Range: -251 to 250.99 %

Steps:

1) Read Total Voltage Bias Percent from Cat ECS controller and convert it to scaled value
2) Read Desired Genset Output Voltage from Cat ECS controller and convert it to scaled value
3) Calculate voltage bias increment command write value

One (1) Word Raw Value (WRITE) = (Engineering Value - Offset) / Scaling

Voltage Bias Percent Increment Command = (Engineering Value – (-251)) / 0.0078125

See following examples for demonstration purposes.

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Example 1 - Voltage Bias Increment Command 0%:

Genset is running at 480 volts. Modbus TCP client sends 0% voltage bias.

Description	Register (Cat ECS 100)	Register Value	Scaled Value
READ - Total Voltage Bias Percent	1452	32128	Register Value x Scaling + Offset (32128 x 0.0078125) + (-251)) = 0%
READ - Desired Genset Output Voltage	1447	480	Register Value x Scaling + Offset (480 x 1) + 0 = 480 V
WRITE - Voltage Bias Percent Increment Command	3	**32128**	Scaled Value = 0 (0%) (Scaled Value - Offset) / Scaling (0 – (-251)) / 0.0078125 = **32128**

Voltage Bias Percent Increment Command value of 32128 to Cat ECS 100 controller will correspond to 0% voltage bias. This example is shown for demonstration purpose; however, there is no need for Modbus client to send a voltage bias increment command if no voltage bias (0%) is desired.

Example 2a - Voltage Bias Increment Command -2%:

Genset is running at 490 volts with an active total voltage bias of 2% (490-480/480 x 100). Modbus TCP client requests Cat ECS over the SCADA Modbus to decrease the voltage by 2% by sending a voltage bias of -2% to decrease the voltage from 490 volts to 480 volts.

Description	Register (Cat ECS 100)	Register Value	Scaled Value
READ - Total Voltage Bias Percent	1452	32384	Register Value x Scaling + Offset (32384 x 0.0078125) + (-251)) = 2%
READ - Desired Genset Output Voltage	1447	490	Register Value x Scaling + Offset (490 x 1) + 0 = 490 V
WRITE - Voltage Bias Percent Increment Command	3	**31872**	Scaled Value = -2 (-2%) (Scaled Value - Offset) / Scaling (-2 - (-251)) / 0.0078125 = **31872**

Example 2b - Voltage Bias Increment Command +2%:

Genset is running at 470 volts with an active total voltage bias of -2% (470-480/480 x 100). Modbus TCP client requests Cat ECS over the SCADA Modbus to increase the voltage by 2% from 470 volts to 480 volts by sending a voltage bias of +2% to increase the voltage from 470 volts to 480 volts.

Description	Register (Cat ECS 100)	Register Value	Scaled Value
READ - Total Voltage Bias Percent	1452	31872	Register Value x Scaling + Offset (31872 x 0.0078125) + (-251)) = -2%
READ - Desired Genset Output Voltage	1447	470	Register Value x Scaling + Offset (470 x 1) + 0 = 470 V
WRITE - Voltage Bias Percent Increment Command	3	**32384**	Scaled Value = 2 (+2%) (Scaled Value - Offset) / Scaling (2 – (-251)) / 0.0078125 = **32384**

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12.4 Base Load (kW) Monitor and Control

The purpose of Base Load Control is to allow the controller to drive the generator set to generate a fixed amount of KW regardless of load characteristics. No Export protection is provided. Consideration for disconnection from Utility in the event of a loss of Power is a critical consideration. An independent Utility availability device should be considered. The unit will not participate in LSLD while in this mode.

- Base Load Control may be active alongside VAR or PF modes.
- Base Load Control is allowed to have a positive value (or zero).
- Base Load Control will automatically be disabled until the controller detects that the gen is connected to Utility (Utility breaker status = closed and Generator breaker status = closed).

This control mode is only active if any Digital Input programmed for “Utility Breaker Aux A” is Active, and AMF is disabled. Therefore, it can be disabled by enabling AMF, or by not programming any Digital Inputs for “Utility Breaker Aux A”.

It is recommended to read the Cat ECS 200 software operating manual section “Base Low (kW) Control” to understand the overall aspect of Base Load control and associated parameters and setpoints.
For SCADA Modbus Base Load control following Source Configuration setpoint must be to configure to SCADA.

Generator Real Power Control Base Load Level Source Configuration
Determines where the controller should look to know the value of the Base Load to which it should regulate. Can be set to any one of the following:

- Setpoint
- Analog Input
- SCADA

Description TCP Port: 50200	Word Count	Register 200P	Scaling, Offset, Range / Status / State / Info
Desired Base Load Command (Read/Write)	2	9 10	Scaling: 1 W / bit Offset: -2000000000 W Range: -2000000000 to +2211081215 W

Reading two (2) word data from Cat ECS two (2) 16-bit consecutive words into DUINT (Double Unsigned Integer) require the following steps:

1) Compute DUINT (Double Unsigned Integer) Value - Left shift MSW Word Register Value (**<<**) 16 bits then Add (**+**) LSW Register Value to it
2) Calculate engineering value using DUINT value, Scaling Factor and Offset

Description TCP Port: 50200	Word Count	Register 200P	Raw Value	Calculation
READ - Desired Base Load Command	2	9 10	30523 56769	**DUINT Value: MSW << 16 + LSW** (30523) **<<** 16 **+** 56769 = 2000412097 **Scaled Value: DUINT Value x Scaling + Offset** 2000412097 x (1) + (-2000000000) = 412097 W

Scaled/Engineering Value: 412097 W (412097 /1000 = 412 kW)

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Writing two (2) word data “DUINT-Double Unsigned Integer” to Cat ECS two 16-bit consecutive words require the following steps:

1) Compute DUINT write value using Desired Base Load engineering value in Watts, Offset and Scaling
2) Compute MSW 16-word write value - Right Shift (**>>**) 16 bits the computed DUINT value
3) Compute LSW 16-word write value - Perform an **AND** operation to DUINT value with 65535 (0xFFFF)
4) Send Write command to the Cat ECS paralleling controller by placing MSW value (30523) at the lower Modbus register address (9) and LSW value (56769) at the higher Modbus Register address (10).

Description TCP Port: 50200	Word Count	Register 200P	Raw Value	Calculation
WRITE - Desired Base Load Command	2	9 10	30523 56769	**Desired Base Load (Engineering Value):** 412097 W **DUINT Write Value:** (Engineering Value - Offset) / Scaling **DUINT Write Value =** ((412097 – (-2000000000)) / 1 **MSW write value: DUINT Value >> 16** (2000412097) **>>** 16 = 30523 **LSW write value: DUINT Value AND 65535** (2000412097) **AND** 65535 = 56769

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12.5 Analog Inputs and Outputs

12.5.1 Read - Analog Input and Output

This example illustrates reading Analog Input #2 value from the Cat ECS 200 Paralleling controller over SCADA Modbus. Use the same steps to read the analog output from Cat ECS controller over SCADA Modbus. In this example, the applied voltage to the analog input #2 is 4 volts.

Figure 12.5.1: Cat ET – Analog Input Configuration

Analog Input #2 Signal Type	Voltage
Analog Input #2 Signal Range	0 to 5 V
Analog Input #2 Configuration Code	Map #1
Analog Input #2 Data Identification	Data Link Only
Analog Input #2 Minimum Data Range	0.000
Analog Input #2 Maximum Data Range	100.000

Description TCP Port: 502	Word Count	Register	Raw Value	Calculation
Cat ECS 200 Paralleling Controller - Analog Input #2 Sensor Value	1	703	320	**Register Raw Value x Scaling + Offset** 320 x (1/16) + (-16) = **4 Volts**
Cat ECS 200 Paralleling Controller - Analog Input #2 Data Value	2	707 708	32283 28170	**DUINT Value: MSW << 16 + LSW** (32283) << 16 = 2115698688 2115698688 + 28170 = 2115726858 **Scaled Value: DUINT Value x Scaling + Offset** 2115726858 x (1/256) + (-8224768) = 39790

12.5.2 Write - Analog Output

In this example, Cat ECS 200 controller Analog Output #2 is configured to 0 to 5V. Cat ECS Controller supports various analog output signal types and ranges; same steps can be applied to evaluate the equation for the different configurations.

Steps:

1) Obtain Analog Output configuration parameters information from Cat ET shown below
   - x1 = Analog Output Minimum Range
   - x2 = Analog Output Maximum Range
   - d1 = Analog Output Minimum Data Range
   - d2 = Analog Output Maximum Data Range
2) Evaluate Analog Output Write equation (Line-Slope) using analog output channel configurations
3) Enter desired analog output value into the equation in volts to calculate the DUINT (Double Unsigned Integer) value

For example, user wants to generator 4.75 volts off analog output #2:

4) Evaluate two-word write value using the equation

Y(x) = mx + c = 5120 (x) + 2105540608

Y (4.75 Volts) = (5120 x 4.75) + 2105540608 = 2105564928

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5) Compute MSW 16-word write value - Right Shift (**>>**) 16 bits using DUINT value (**2105564928)**
   - **MSW Word Value = 32128**
6) Compute LSW 16-word write value - Perform an **AND** operation to DUINT value (**2105564928)** with 65535 (0xFFFF)
   - **LSW Word Value = 24320**
7) Send Write command to the Cat ECS Paralleling controller (TPC Port 50200) by placing MSW value "32128" at the lower Modbus register address (19) and LSW value "24320" at the higher Modbus Register address (20)

Figure 12.5.2: Cat ET – Analog Output Configuration

Analog Output #2 Output Type	Voltage
Analog Output #2 Output Range	0 to 5 V
Analog Output #2 Data Identification	Data Link Only
Analog Output #2 Minimum Data Range	0.000
Analog Output #2 Maximum Data Range	100.000

Table 12.5.2: Illustration – Analog Output Write Equation Evaluation

OS - Offset	-8224768
SF - Scaling	1/256
y1 – Minimum Value **d1 = 0**	***y1 = (d1-OS) / SF*** *y1 = (0 - (-8224768))* x 256 **y1 = 2105540608**
Y2 – Maximum Value **d2 = 100**	***y2 = (d2-OS) / SF*** *y2 = (100 - (-8224768))* x 256 **y2 = 2105566208**
Slope (m) **x1 = 0 volts** **x2 = 5 volts**	**m = (y2-y1)/(x2-x1)** = (2105566208 – 2105540608) / (5-0) **m = 5120**
y-Intercept = c **x1 = 0 volts**	y - y1 = m(x-x1) y = m (x - x1) + y1 = 5120 (0 – 0) + 2105540608 = 2105540608 **c = 2105540608**
Analog Output Write	**Y(x) = mx + c = 5120 (x) + 2105540608**

Description TCP Port: 50200	Word Count	Modbus Register		
		100G	200G	200P
Analog Output #2	2	35	17	19

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12.5.3 Read - PWM Output

This example illustrates reading PWM Output #2 value from Cat ECS Paralleling controller over the SCADA Modbus. In this example, the PWM Output #2 is 25%.

Figure 12.5.3: Cat ET – PWM Output Configuration

PWM Output #2 Enable Status	Enabled
PWM Output #2 Output Range	0 to 100%
PWM Output #2 Data Identification	Data Link Only
PWM Output #2 Minimum Data Range	0.000
PWM Output #2 Maximum Data Range	100.000

Description TCP Port: 502	Word Count	Register	Raw Value	Calculation
PWM Output #2 Output Value	1	724	10000	**Register Raw Value x Scaling + Offset** 10000 x (0.0025) + 0 = 25 **%**
PWM Output #2 Data Value	2	719 720	32128 6656	**DUINT Value: MSW << 16 + LSW** (32128) << 16 = 2105540608 2105540608 + 6656 = 2105547264 **Scaled Value: DUINT Value x Scaling + Offset** 2105547264 x (1/256) + (-8224768) = 26

12.5.4 Write - PWM Output

This example illustrates writing to Cat ECS Paralleling controller Analog PWM Output #2. Generating the desired PWM out signal of Cat ECS controller requires the following steps:

Steps:

1) Obtain PWM Output configuration parameters information from Cat ET shown below
   - x1 = PWM Output Minimum Range
   - x2 = PWM Output Maximum Range
   - d1 = PWM Output Minimum Data Range
   - d2 = PWM Output Maximum Data Range

2) Evaluate PWM Output Write equation (Line-Slope) using PWM output channel configurations
3) Enter desired PWM output value into the equation in % to calculate the DUINT (Double Unsigned Integer) value

For example, user wants to generator 75% PWM output:

4) Evaluate two-word write value using the equation

$$Y(x) = mx + c = 256\ (x) + 2105540864$$

$$Y\ (75\%) = (256 \times 75) + 2105540864 = 2105560064$$

5) Compute MSW 16-word write value - Right Shift (**>>**) 16 bits using DUINT value (**2105560064)**
   - **MSW Word Value = 32128**
6) Compute LSW 16-word write value - Perform an **AND** operation to DUINT value (**2105560064)** with 65535 (0xFFFF)
   - **LSW Word Value = 19456**

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7) Send Write command to the Cat ECS Paralleling controller (TPC Port 50200) by placing MSW value “32128” at the lower Modbus register address (23) and LSW value “19456” at the higher Modbus Register address (24)

Figure 12.5.4: Cat ET – PWM Output Configuration

PWM Output #2 Enable Status	Enabled
PWM Output #2 Output Range	0 to 100%
PWM Output #2 Data Identification	Data Link Only
PWM Output #2 Minimum Data Range	0.000
PWM Output #2 Maximum Data Range	100.000

Table 12.5.4 Illustration – PWM Output Write Equation Evaluation

OS - Offset	-8224768
SF - Scaling	1/256
y1 – Minimum Value **d1 = 0**	***y1 = (d1-OS) / SF*** *y1 = (0 - (-8224768))* x 256 **y1 = 2105540608**
y2 – Maximum Value **d2 = 100**	*y2 = (d2-OS) / SF* *y2 = (100 - (-8224768))* x 256 **y2 = 2105566208**
Slope (m) **x1 = 0%** **x2 = 100%**	**m = (y2-y1)/(x2-x1)** = (2105566208 – 2105540608) / (100-0) **m = 256**
Y-Intercept = c **x1 = 0%**	y - y1 = m(x-x1) y = m (x - x1) + y1 = 256 (0 – 0) + 2105540608 = 2105540864 **c = 2105540864**
PWM Output Two-Word WRITE Value	**Y(x) = mx + c = 256 (x) + 2105540864**

Description TCP Port: 50200	Word Count	Modbus Register		
		200G	200G	200P
PWM Output #2	2	39	21	23

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12.6 Remote Analog Inputs and Outputs

12.6.1 Read - Remote Analog Input and Output

This example illustrates reading Remote Analog Input #1 (4 to 20mA) from the Cat ECS 200 Paralleling controller over SCADA Modbus. Use the same steps to read the remote analog output from Cat ECS controller over SCADA Modbus. In this example, the applied voltage to the analog input #1 is 12mA.

Figure 12.6.1: Cat ET – Remote Analog Input Configuration

Description	Value
Remote Analog Inputs	
Remote Analog Input #1 Signal Type	Current
Remote Analog Input #1 Data Identification	Data Link Only
Remote Analog Input #1 Minimum Data Range	0.000
Remote Analog Input #1 Maximum Data Range	100.000

Description TCP Port: 502	Word Count	Register	Raw Value	Calculation
Cat ECS 200 Paralleling Controller – Remote Analog Input #1 Sensor Value	1	1627	448	**Register Raw Value x Scaling + Offset** 448 x (1/16) + (-16) = **12mA**
Cat ECS 200 Paralleling Controller – Remote Analog Input #1 Data Value	2	1635 1636	MSW LSW	**DUINT Value: MSW << 16 + LSW** **Scaled Value: DUINT Value x Scaling + Offset** **=** DUINT x (1/256) + (-8224768)

12.6.2 Write - Remote Analog Output

This example illustrates writing to Remote Analog Output #1 (4 to 20mA) connected to Cat ECS Paralleling controller.

Steps:

1) Obtain Remote Analog Output configuration parameters information from Cat ET shown below
   - d1 = Analog Output Minimum Data Range
   - d2 = Analog Output Maximum Data Range
2) Evaluate Analog Output Write equation (Line-Slope) using remote analog output channel configurations
3) Enter desired analog output value into the equation in volts to calculate the DUINT (Double Unsigned Integer) value

For example, user wants to generator 12mA volts off remote analog output #1:

4) Evaluate two-word write value using the equation

$$Y(x) = mx + c = 1600\ (x) + 2105534208$$

$$Y\ (12mA) = (1600 \times 12) + 2105534208 = 2105553408$$

5) Compute MSW 16-word write value - Right Shift (**>>**) 16 bits using DUINT value (**2105553408)**
   - **MSW Word Value = 32128**

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6) Compute LSW 16-word write value - Perform an **AND** operation to DUINT value (**2105553408)** with 65535 (0xFFFF)
   - **LSW Word Value = 12800**
7) Send Write command to the Cat ECS Paralleling controller (TPC Port 50200) by placing MSW value "32128" at the lower Modbus register address (87) and LSW value "12800" at the higher Modbus Register address (88)

Figure 11.1 Cat ET – Remote Analog Output Configuration

Description	Value
Remote Analog Outputs	
Remote Analog Output #1 Output Type	Current
Remote Analog Output #1 Data Identification	Data Link Only
Remote Analog Output #1 Minimum Data Range	0.000
Remote Analog Output #1 Maximum Data Range	100.000

OS - Offset	-8224768
SF - Scaling	1/256
y1 – Minimum Value **d1 = 0**	***y1 = (d1-OS) / SF*** *y1 = (0 - (*-8224768)) x 256 **y1 = 2105540608**
y2 – Maximum Value **y2 = 100**	***y2 = (d2-OS) / SF*** *y2 = (100 - (*-8224768)) x 256 **y2 = 2105566208**
Slope (m) **x1 = 4mA** **x2 = 20mA**	**m = (y2-y1)/(x2-x1)** = (2105566208 – 2105540608) / (20-4) **m = 1600**
Y-Intercept = c **x1 = 0**	y - y1 = m(x-x1) y = **m (x - x1) + y1** = 1600 (0 – 4) + 2105540608 = 2105534208 **c = 2105534208**
Remote Analog Output Two-Word Write	**Y(x) = mx + c = 1600 (x) + 2105534208**

Description TCP Port: 50200	Word Count	Modbus Register		
		100G	200G	200P
Remote Analog Output #1	2	91	85	87

12.6.3 Read - Remote RTD Input

This example illustrates reading Remote RTD Input #1 value from the Ethernet IO module connected to Cat ECS Paralleling controller.

Description	Word Count	Register	Raw Value	Calculation
Remote RTD Input #1 Sensor Value	1	1675	500	**Register Raw Value x Scaling + Offset** 500 x (0.1) + (200) = **250 Deg C**

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12.7 Event Logs

The Cat ECS 100 and Cat ECS 200 Event Logs can be viewed via Cat ET (Caterpillar Service Tool) by selecting the Logged Diagnostic Codes button on the toolbar or the HMI display under event logs menu. There are up to forty (40) Event Log Entries supported in the Cat ECS controller per installed module, and each Log Entry consists of 14 Modbus Registers. It is recommended to consult HMI and Cat ECS software operating manuals for further information on Event Logs and Diagnostics.

For user demonstration purposes, Genset and Engine Event Log Entry #1 and Entry #40 registers are shown below. Refer to Cat ECS 100 v1.4 and Cat ECS 200 v1.1 SCADA register lists spreadsheets for all other register definitions.

Description TCP Port: 50201	Word Count	Modbus Register	
		100G	200G
Genset Event Log Entry #1 to Genset Event Log Entry #40			
Genset Event Log Entry 1	14	51	51
Genset Event Log Entry 40	14	597	597
Engine Event Log Entry #1 to Engine Event Log Entry #40			
Engine Event Log Entry 1	14	2650	2650
Engine Event Log Entry 40	14	4879	4879

Example: Engine Event Log – Entry #1 (Generator Control Not in Automatic Warning)

In this example, Generator Control Not in Automatic Warning is the Event Log Entry #1 under Engine Event Logs. The Modbus Registers for Event Log Entry #1 are from 2650 to 2663. The format of these fourteen (14) registers (2650 to 2663) and their associated details are shown below.

- **Event Log Timestamp (MM/DD/YYYY HH:MM:SS AM/PM):**
  - **Last Timestamp:** Register 2650 to 2652
  - **First Timestamp:** Register 2653 to 2655
- **Event Log Hour Meter (seconds)**
  - **Last Hourmeter:** Register 2656 to 2657
  - **First Hourmeter:** Register 2658 to 2659
- **Event Log Flag (Occurrence, Event Status, Lamp Status):** Register 2660
- **Event Log SPN-FMI (Event ID):** Register 2661 to2662
- **Event Log Entry Index Number (J1939 Count):** Register 2663

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Figure: 12.7 – HMI Display Event Log

1) Event Identifier
2) SPN #
3) FMI #
4) Event Status
5) Lamp Status
6) First Timestamp
7) First Hourmeter
8) Last Timestamp
9) Last Hourmeter
10) Occurrence
11) Number of Events in Engine Events Log

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Event Log Timestamp:

Byte	Resolution	Offset	Data Range
5	1 year / bit	1985 years	1985 to 2235 years
4	0.25 days / bit	0.75 days (12:00-6:00am on day 1 = 0x01)	0 to 62.5 days
3	1 month / bit	1 month (Jan = 0x01)	0 to 250 months
2	1 hour / bit	0 hours (midnight = 0x00)	0 to 250 hours
1	1 minute / bit	0 minutes (HH:00 = 0x00)	0 to 250 minutes
0	0.25 seconds / bit	0 seconds (HH:MM:00 = 0x00)	0 to 62.5 seconds

Event Log Hourmeter:

Resolution	Offset	Data Range
1 seconds / bit	0 seconds	0 to 4294967294 seconds

Event Log Flag (Occurrence, Event Status and Lamp Status)

- **Bit-0 to Bit-7: Number of Event Occurrences Count**
- **Bit-8 to Bit-11: Event Status**
  - 0 (0000_b) = Inactive
  - 4 (0100_b) = Active
  - 5 (0101_b) = Present
  - 15 (1111_b) = Unavailable

Note: If the Event Status value is other than 0, 4, 5 or 15 then the HMI hides the value (Blank Space).

- **Bit-12 to Bit-14: Lamp Status (see Table 11.21 below)**
  - 0 (000_b) = Protect Lamp
  - 1 (001_b) = Amber Lamp
  - 2 (010_b) = Red Lamp
  - 3 (011_b) = Malfunction Lamp
  - 4 (111_b) = No lamp status

Lamp Status	Description	HMI Symbol
Protect Lamp	HMI only shows the '**Amber Triangle**' warning indication symbol.	
Amber Lamp	HMI only shows the '**Amber Triangle**' warning indication symbol.	
Red Lamp	HMI only shows the '**Red Octagon**' shutdown indication symbol.	
Malfunction Lamp	HMI only shows the '**Amber Triangle**' warning indication symbol.	
No lamp status	If the bit value combination is '**111**' (special case), the controller shall handle the logic that defines the lamp status based on the FMI code and populate **Bits 14, 13** and **12**. The HMI reads the bits and hide the Lamp Status shutdown and warning symbols.	N/A

Note: For 'non-valid' bit value combinations, the HMI hides the lamp status shutdown and warning symbols. In case of non-valid bits or '111' (No Lamp Status) the event status is displayed as "Unavailable". Bit

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combination '**111**' and **non-valid** bit combinations are processed the same to either not display the Shutdown and Warning icons (Active/Logged screen) or show 'Unavailable' (Event Detail screen).
Bit-15: Unused

SPN-FMI

- **FMI / WCI:** Bit-0 to Bit-4
- **SPN / CID / EID:** Bit-5 to Bit-23
- **Event Identifier Type:** Bit-24 to Bit-25
  - 0 (00_b) = CID
  - 1 (01_b) = EID
  - 2 (10_b) = Unused
  - 3 (11_b) = SPN
- **Unused:** Bit-26 to Bit-31

Event Log Entry Index Number

The Log Entry Index represents the total number of available events in the module event log.

Resolution	Offset	Data Range
1 / bit	0	0 to 64255

Event Log Consist Summary

Engine Event Log – Entry #1 (Generator Control Not in Automatic Warning)		
Register Index	**Modbus Register**	**Description and Values**
0	2650	**Last Timestamp :** 01/16/2007 09:16:44 AM
1	2651	
2	2652	
3	2653	**First Timestamp:** 11/11/1984 00:00:00 AM
4	2654	
5	2655	
6	2656	**Last Hourmeter:** 0 Hours
7	2657	
8	2658	**First Hourmeter:** 0 Hours
9	2659	
10	2660	**Occurrence:** 9 (00001001_b) **Event Status:** Present (0101_b) **Lamp Status:** Amber Lamp (001_b)
11	2661	**Event Identifier:** 3 **SPN:** 4007 **FMI:** 31
12	2662	
13	2663	**Log Entry Index:** 2

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13.0 Best Practices

Best practices in Modbus communication are crucial for reliable, efficient, and secure data exchange between devices in industrial control systems. Adhering to these practices not only minimize communication intermittences and failures but help achieving predictable performance and enhance the overall system performance.

13.1 Data and Network Optimization

- **Batch Read Operations** - Minimize number of requests by reading multiple registers in a single transaction instead of making individual requests for each register.
- **Data Prioritization and Selective Polling -** Identify and prioritize critical data that needs to be read/written more frequently. Poll critical registers more frequently and less critical registers less often by adjusting polling rate to a frequency that balances between real-time data needs and network load. Use event-driven updates where possible.
- **Network Congestion & Latency -** Be aware of network latency. If you have a high-latency connection, you might need to increase timeouts. Don't create a new connection for every Modbus transaction.
- **Monitor Communication Performance** - Track the response times of Modbus requests and identify any delays or bottlenecks. Use logging to record communication errors and performance metrics in your software.

13.2 Retry Mechanisms and Error Handling

- **Retry Mechanism** - Implement a retry mechanism for failed transactions but limit the number of retries to avoid network congestion. Use timeouts to detect unresponsive devices and avoid hanging the application. Too short a timeout can cause unnecessary retries; too long may result in slow responses.
- **Error Handling - Diagnostics and Logging** - When errors occur, properly interpret the Modbus exception codes (e.g., illegal function, illegal data address) and handle retries or fallbacks. Log communication errors and exceptions for debugging and troubleshooting. See troubleshooting section below for further information on this topic.

13.3 Security Considerations

- **Secure Modbus TCP Communication** - Consider implementing network security measures such as VPNs or firewalls to protect against unauthorized access.
- **Log and Validate Input Data** – Log and validate critical data reads and writes to prevent invalid or malicious data from affecting your system.

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14.0 Troubleshooting

14.1 Intermittent or Failed Communication

There are several factors which can cause intermittent or failed communication. Some may cause a SCADA Data Link Fault event on the Cat ECS controller:

- Incorrect or bad wiring
- Poor shielding or no shielding (especially in noisy environments like adjacent to AC power cables or near the generator housing), or low-spec cable (anything below Cat-5)
- Exceeding Ethernet cable length limit of 100m (325ft)
- Ethernet cable disconnected after communication was established
- Bad IP address or IP address not unique
- Subnet mask or gateway mismatch

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14.2 Modbus Errors - Exception Responses

The Cat ECS controller sends an Exception response if it cannot interpret the request due to various reasons. The exception response is identified by a function code that is 128_d (0x80) or greater. There is additional data within the exception response message that can help identify the problem. The different exception responses and exception codes are described below.

131 (0X83) – EXCEPTION RESPONSE TO READ REGISTERS QUERY

The exception response contains a 1-byte exception code.

Code	Reason
01	Starting or ending (starting + count) register address invalid
02	Register Count was less than 1 or greater than 125
03	Read error possibly due to some register(s) in the span not being Read registers

134 (0X86) – EXCEPTION RESPONSE TO WRITE SINGLE REGISTER QUERY

The exception response contains a 1-byte exception code.

Code	Reason
02	Register address invalid
03	Register value out of range perhaps because longer than 2 bytes
04	Write error possibly due to not being a Write register

144 (0X90) – EXCEPTION RESPONSE TO WRITE MULTIPLE REGISTERS QUERY

The exception response contains a 1-byte exception code.

Code	Reason
02	Starting or ending (starting + count) register address invalid
03	Register Count was less than 1 or greater than 123, or byte count was not (Register Count x 2)
04	Write error possibly due to some register(s) in the span not being Write registers

OTHER 128 AND ABOVE (0X80 AND ABOVE) – EXCEPTION RESPONSE TO UNSUPPORTED QUERIES

The ECS does not support any function codes not listed above in the Supported Function Codes Chapter. Such queries will return an exception response with this 1-byte exception code.

Code	Reason
01	Unsupported function code

Figure 14.2: Modbus Errors – Exception Responses

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14.3 Invalid or Out of Range Data

Modbus TCP client must interpret that data correctly. For each piece of data, there is a certain range of data that is defined as valid data shown in the data tables as Range.

If the data is not within the defined valid data range, then it could be because of several possible reasons:

1. The data is expected over either the Primary or Accessory Data Link, and is not being received
2. The sensing inputs providing the data detect a fault (e.g., open circuit, short circuit)
3. The Cat ECS has determined a logical flaw with the data being detected (e.g., generator circuit breaker aux A and aux B either both closed or both open)
4. The data is expected from an input, but input is not configured to provide the data
5. The data is expected from Primary or Accessory Data Link, and is being received as faulty due to problems with the module or the connection

Following troubleshooting steps can be followed to investigate invalid data:

1. Check whether the feature being tested is supported on this package
2. Check the definition of the data in this document, and determine whether the data may be unavailable due to certain conditions not having been met (e.g. cooldown duration remaining is only available while cooldown is underway)
3. Confirm that the data is truly invalid by:
    a. If the same data is available on Cat ECS display or Cat ET Data Status Monitor. Invalid data should show up as asterisks (****)
    b. Using Modbus Simulation tools by Known Third Party
4. Once the invalid data is confirmed, determine the source of the data; the definition of that data in this document will help. Sometimes there are multiple sources (e.g., engine speed can come from either the engine ECM over the Primary Data Link, or from a magnetic pickup sensor, and the selected source must be configured in Cat ECS.
5. Once the source is determined, troubleshoot the sensor input(s) that generates this data:
    a. Poor or incorrect wiring, or signal reference
    b. Mismatched sensor type (either plugging the sensor in to the wrong input, or configuring the input for a different sensor type
6. Try disconnecting the sensor from the ECM and wiring separately to the sensor, or even to a duplicate sensor that is not connected into the sensor port
7. Try a new sensor, or one that is known to be working
8. If the sensor can be connected to multiple places, try connecting to a different place (such as a configurable digital or analog input on the Cat ECS), temporarily disabling the normal setting; if that works, it may indicate a hardware fault with the sensing circuit

If none of these steps resolve the issue, contact your Caterpillar technical support representative.

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